
MTH2203:

NUMERICAL ANALYSIS I

LECTURE NOTES

PREPARED BY

G. KAKUBA
Department of Mathematics
Makerere University



FEBRUARY 18, 2019

Preface

These lecture notes are based on the Numerical Analysis I course outline as described in the latest (April 2013) Undergraduate Handbook of the Department of Mathematics at Makerere University. It is a second year course of three credit units.

Numerical Analysis is a subject at the heart of scientific computing and thus plays a critical role in the development of computer solutions for mathematical problems.

According to the undergraduate handbook, by the end of the course, the student should be able to use numerical techniques to solve and analyse solutions of problems in interpolation, numerical integration and differentiation, non-linear equations and systems of linear equations. The students should be able to write codes for simple numerical analysis algorithms for these solutions.

Students are expected to have gone through successfully the first year courses of linear algebra and calculus. Some basic knowledge of computer packages such as Matlab, Maple or Fortran will be helpful.

The course is a very practical course and is best learnt by doing. Students are strongly encouraged to do a lot of exercises outside the class. The course mainly involves finding numerical solutions for mathematical problems but what is even more critical is how accurate and efficient the solutions and the methods used are.

Appreciation to my former Numerical Analysis I students Galiwango Ronald (currently a PhD student at the University of Georgia, USA) and Katende Ronald (Just completed his MSc at Pan African University in Nairobi) for assisting in putting the lecture notes in $\text{\LaTeX}$.

©Department of Mathematics, P. O. Box 7062, Kampala

godwin kakuba
BSc(M,P,Ed)-Mak, MSc(CSE)-TU/e, PhD-TU/e
godwin.a.kakuba@gmail.com

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1 Introduction

Numerical Analysis is a subject at the heart of scientific computing and plays a critical role in the development of computer solutions for mathematical problems. It concerns itself with the study and analysis of computer algorithms for solving problems that arise in science. The subject covers the approximation of functions, approximation of integrals, approximations of solutions of algebraic, transcendental, differential and integral equations.

The course Numerical Analysis I introduces students to the science of approximating solutions to problems of continuous mathematics, providing alternatives and sometimes the only preferred methods to problem solving. A good numerical method is one that tells you how good the approximation is and/or how fast a sequence of convergent approximations can be generated. Therefore particular emphasis should be put on the stability, accuracy, efficiency and reliability of numerical algorithms. It is fundamental to try to avoid the risk described by Zarowski in [?] as “adopting to a “grocery list” or “recipes” approach (i.e., “methods” at the expense of “analysis”) wherein several methods are presented, but with little serious discussion of issues such as how they are obtained and their relative advantages and disadvantages, and with little consideration to error analysis, convergence properties, or stability issues.” This will pose a disadvantage to the student on the future application of the methods or further studies. Important, therefore, would be to at least tell users if a particular algorithm will converge or how good the estimates are compared to the exact values, the ones that would be obtained using analytical approaches. This goodness of results, the accuracy, is best gauged by the *error*. Absolute error is the difference between the estimate and the exact value. Relative error is the ratio of absolute error to the exact value. Several things might play into the accuracy of any computation. Accuracy is affected by *error in the inputs*, *round off errors* and *approximation errors*. Round off errors are due to calculations that use numbers that can only be represented using a finite number of digits. Approximation error can be due various reasons, such as truncation and discretisation. *Truncation error* is due to approximating an infinite series by a few finite terms of the series. *Discretization error* is due to replacing a continuous quantity by a discrete one, for example a differential quotient may be replaced by a difference quotient. If the algorithm involves several steps, then the error at each step is called the *local error* and the error due to the cumulative effect of the errors at all the steps is the *global error*. This error, the global error, is what we actually pass on to the user of the results. If an error, such as the error in inputs, does not amplify further as the computation continues, but rather dampens, then we say we have numerical stability.

The course Numerical Analysis I will discuss these things without going out of context of a 3CU undergraduate course. This course is also a prerequisite for a build on course in third year, Numerical Analysis II, that is recommended in order for you to appreciate the dose that the university is prepared to give you in this area of mathematics.

2 Interpolation

Interpolation simply means fitting some function to given data so that the function has the same values as the given data. The simplest interpolation problem in one dimension is of the following form. For given data $(x_k, f_k), k = 0, 1, \dots, n$, we seek a function $p_n(x)$ such that $p_n(x_k) = f_k$. The function $p_n(x)$ is called an *interpolation function* or an *interpolant* for the given data.

Definition 2.1 The process of calculating $f(x)$ for $x \in [x_0, x_n]$ and $x \neq x_k, k = 0, 1, 2, \dots, n$, is called *interpolation*.

2.1 Purposes of interpolation

Some of the purposes of interpolation include:

1. Estimating a function that is otherwise known only at discrete points.
2. Plotting a smooth curve through discrete points.
3. Differentiating or integrating discrete data.
4. Understanding a given data set.
5. Replacing a “difficult” function with an “easy” one.
6. a basis for development of quadrature rules and solutions of ODEs.

2.2 Finding the interpolant

The selection of the interpolant is usually based on how easy the function is to work with, that is, determining its parameters from data, evaluating the function at a given point, differentiating or integrating the function, and, how well the properties of the function match the properties of the data to be fit, that is, smoothness, monotonicity, convexity, periodicity, among others. Some families of functions commonly used for interpolation include: *Polynomials, Piecewise polynomials, Trigonometric functions, Exponentials and Rational functions*.

The process of finding the interpolant begins with deciding on what family of interpolants to use by choosing the basis. The family of functions chosen for interpolating a given set of data points is spanned by a set of *basis functions*

$$\{\phi_0(x), \phi_1(x), \dots, \phi_n(x)\}.$$

The interpolating function is then chosen as a linear combination of these basis functions, that is,

$$p_n(x) = \sum_{j=0}^n \alpha_j \phi_j(x) \tag{1}$$

where the parameters α_j are to be determined. Now $p_n(x)$ is required to interpolate (i.e. go through) the data points (x_k, f_k) , $k = 0, 1, 2, \dots, n$. This means that

$$p(x_k) = \sum_{j=0}^n \alpha_j \phi_j(x_k) = f_k \quad \text{for } k = 0, 1, \dots, n,$$

which is a system of $n + 1$ equations in $n + 1$ unknowns $\alpha_0, \alpha_1, \dots, \alpha_n$. In this course we will discuss polynomial interpolation in which the basis functions are polynomials. These may be the monomials, Newton polynomials, Lagrange polynomials, orthogonal polynomials, etc.

2.2.1 Lagrange Interpolation

Let us consider the case of interpolating a function $f(x)$ which is continuous on $[a, b] \subset \mathbb{R}$. Let $f(x_k) = f_k$ for $k = 0, 1, \dots, n$ with $x_k \in [a, b]$. We seek a polynomial $p_n(x)$ of degree n such that $p_n(x_k) = f_k$, $k = 0, 1, \dots, n$. We will also use the following notation: Let $U_k : k = 0, 1, \dots, n$ be any set of numbers, we define the following products

$$\prod_{k=0}^n U_k = U_0 U_1 U_2 \cdots U_n \quad \text{and} \quad \prod_{k=0, k \neq j}^n U_k = U_0 U_1 \cdots U_{j-1} U_{j+1} \cdots U_n$$

For a given set of $(n + 1)$ points, the Lagrange basis polynomials are given by

$$L_k(x) = \prod_{j=0, j \neq k}^n \left(\frac{x - x_j}{x_k - x_j} \right). \quad (2)$$

Now the interpolant is given by

$$p_n(x) = \sum_{j=0}^n \alpha_j \phi_j(x).$$

Then, by the interpolation property $p_n(x_k) = f_k \forall k$ we have that,

$$p_n(x_k) = \sum_{j=0}^n \alpha_j \phi_j(x_k) = f_k, \quad k = 0, 1, 2, \dots, n. \quad (3)$$

So we have the system

$$\begin{aligned}
 \alpha_0 \phi_0(x_0) + \alpha_1 \phi_1(x_0) + \cdots + \alpha_n \phi_n(x_0) &= f_0 \\
 \alpha_0 \phi_0(x_1) + \alpha_1 \phi_1(x_1) + \cdots + \alpha_n \phi_n(x_1) &= f_1 \\
 \alpha_0 \phi_0(x_2) + \alpha_1 \phi_1(x_2) + \cdots + \alpha_n \phi_n(x_2) &= f_2 \\
 \vdots & \\
 \alpha_0 \phi_0(x_n) + \alpha_1 \phi_1(x_n) + \cdots + \alpha_n \phi_n(x_n) &= f_n
 \end{aligned}$$

or

$$\begin{pmatrix} \phi_0(x_0) & \phi_1(x_0) & \cdots & \phi_n(x_0) \\ \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_n(x_1) \\ \vdots & \vdots & & \vdots \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_n(x_n) \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_n \end{pmatrix}$$

that is

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{f} \tag{4}$$

where the matrix $\mathbf{A}$ is such that $a_{kj} = \phi_j(x_k)$, the vectors $\boldsymbol{\alpha}$ and $\mathbf{f}$ are defined as $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \alpha_2, \dots, \alpha_n)^\top$ and $\mathbf{f} = (f_0, f_1, f_2, \dots, f_n)^\top$. Now

$$\phi_j(x) = L_j(x) = \prod_{i=0, i \neq j}^n \left(\frac{x - x_i}{x_j - x_i} \right).$$

Then

$$\phi_j(x_k) = L_j(x_k) = \prod_{i=0, i \neq j}^n \left(\frac{x_k - x_i}{x_j - x_i} \right),$$

so that if $k = j$, we have

$$\phi_j(x_k) = L_j(x_j) = \prod_{i=0, i \neq j}^n \left(\frac{x_j - x_i}{x_j - x_i} \right) = 1$$

For $k \neq j$ we have

$$\phi_j(x_k) = L_j(x_k) = \prod_{i=0, i \neq j}^n \left(\frac{x_k - x_i}{x_j - x_i} \right),$$

and there is some factor for which $i = k$, that is, $x_k - x_k = 0$, so that $\phi_j(x_k) = 0$. In summary we have

$$\phi_j(x_k) = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases}$$

Therefore the coefficient matrix in the system (11) is an identity matrix so that the solution of the system is $\alpha = f$. The coefficients for Lagrange interpolation are therefore equal to the function values. That is, $\alpha_0 = f_0, \alpha_1 = f_1, \dots, \alpha_n = f_n$. Hence

$$p_n(x) = \sum_{k=0}^n f_k L_k(x). \quad (5)$$

Example 2.1 Given that $f(0) = -2, f(2) = 4, f(3) = 10$ is data from some function $f(x)$.

- (i) Find the quadratic interpolant for the data.
- (ii) Hence interpolate f at $x = 1, x = 2.2$ and extrapolate at $x = 4$.

Solution: Let us summarise the given data of Example 2.1 in tabular form

	x_0	x_1	x_2
x_k	0	2	3
$f(x_k)$	-2	4	10
	f_0	f_1	f_2

- (i) Then the basis polynomials are

$$L_0(x) = \prod_{j=0, j \neq 0}^2 \frac{(x - x_j)}{(x_0 - x_j)} = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 2)(x - 3)}{(0 - 2)(0 - 3)} = \frac{1}{6}(x^2 - 5x + 6)$$

$$L_1(x) = \prod_{j=0, j \neq 1}^2 \frac{(x - x_j)}{(x_1 - x_j)} = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{(x - 0)(x - 3)}{(2 - 0)(2 - 3)} = -\frac{1}{2}(x^2 - 3x)$$

$$L_2(x) = \prod_{j=0, j \neq 2}^2 \frac{(x - x_j)}{(x_2 - x_j)} = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{(x - 0)(x - 2)}{(3 - 0)(3 - 2)} = \frac{1}{3}(x^2 - 2x)$$

Therefore,

$$\begin{aligned} p_2(x) &= -2 \cdot \frac{1}{6}(x^2 - 5x + 6) + 4 \cdot \frac{1}{2}(3x - x^2) + 10 \cdot \frac{1}{3}(x^2 - 2x) \\ &= x^2 + x - 2. \end{aligned}$$

(ii) $p_2(x) = x^2 + x - 2$, hence

$$\begin{aligned} f(1) &\simeq p_2(1) = 1^2 + 1 - 2 = 0 \\ f(2.2) &\simeq \\ f(4) &\simeq \end{aligned}$$

Theorem 2.1 (Interpolation Error) *If f is a sufficiently smooth function and p_n is the polynomial of degree at most n that interpolates f at $n + 1$ points $x_0, x_1, x_2, \dots, x_n$, then for each $x \in [x_0, x_n]$ there is a $\xi \in (x_0, x_n)$ such that*

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n). \quad (6)$$

Theorem 2.2 (Interpolant Uniqueness) *For a given set of data (x_k, f_k) , $k = 0, 1, \dots, n$, the interpolating polynomial $p_n(x)$ is unique.*

Proof: By contradiction, let $p_n(x)$ and $q_n(x)$ be different polynomials which interpolate $f(x)$ over the nodes $\{x_k; k = 0, 1, 2, \dots, n\}$ on $[a, b]$. Thus

$$p_n(x_k) = q_n(x_k) = f(x_k), \text{ for } k = 0, 1, 2, \dots, n.$$

Define $r(x) = p_n(x) - q_n(x) \forall x \in [a, b]$. Then $r(x)$ is of degree at most n . But

$$r(x_k) = p_n(x_k) - q_n(x_k) = 0 \text{ for } k = 0, 1, 2, \dots, n.$$

This means that $r(x) = 0$ has $n + 1$ roots, yet $r(x)$ is a polynomial of degree at most n . This is a contradiction to the fundamental law of algebra, i.e., a non zero polynomial of degree n has at most n roots/zeros. Hence $p_n(x)$ is unique.

Exercises

Exercise 2.2.1-1: Construct a linear interpolating polynomial $P(x)$ for the function $f(x) = \frac{1}{x}$ on $[1, 2]$. Use the polynomial to interpolate $f(x)$ at $x = 1.2$ [$P_1(1.2) = 0.9$]

Exercise 2.2.1-2: Given $x_0 = 0$, $x_1 = 1/2$ and $x_2 = 1$. For $f(x) = e^x$, construct a lagrange polynomial that agrees with $f(x)$ at the interpolating points.

Exercise 2.2.1-3: Find a third degree lagrange polynomial that goes through the points $(0,0)$, $(1,1)$, $(8,2)$ and $(27,3)$. Use the polynomial to find α in $(20, \alpha)$. Also construct a linear interpolating polynomial using only $(8,2)$ and $(27,3)$.

Exercise 2.2.1-4: Find the interpolating polynomials going through,

- (i) $(0, 1)$ and $(2, 3)$
- (ii) $(-1, a)$, $(0, b)$ and $(1, c)$.
- (iii) $(0, 1)$, $(1, 0)$, and $(3, 0)$.

Exercise 2.2.1-5: Given the table below, use a lagrange interpolating polynomial of degree one, two and three to find $f(1.5)$.

x	2.0	2.2	2.4	2.6
f(x)	0.5102	0.5108	0.5104	0.4813

Exercise 2.2.1-6: Prove Theorem 2.1.

2.2.2 The monomial basis

The monomial basis functions are given by

$$\phi_j(x) = x^j, j = 0, 1, \dots, n. \quad (7)$$

The interpolating polynomial is again given by

$$p_n(x) = \alpha_0 \phi_0(x) + \alpha_1 \phi_1(x) + \dots + \alpha_n \phi_n(x).$$

Now, $p_n(x_k) = f_k$, means

$$\alpha_0 + \alpha_1 x_k + \alpha_2 x_k^2 + \dots + \alpha_n x_k^n = f_k, \quad k = 0, 1, 2, \dots, n.$$

This results in the system

$$\begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_n \end{pmatrix} \quad (8)$$

which needs to be solved to find the parameters $\alpha_0, \alpha_1, \dots, \alpha_n$. The coefficient matrix in the system (8) is a vandermode matrix

$$\mathbf{A} = \begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ 1 & x_2 & x_2^2 & \cdots & x_2^n \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{pmatrix}.$$

It is a full matrix and therefore the process of finding the parameters α is in this case more expensive than when we use the Lagrange basis. Furthermore, the matrix gets singular in higher order of if the nodes are too close.

2.2.3 The Newton Basis

The Newton basis polynomials are defined as

$$\phi_j(x) = \prod_{i=0}^{j-1} (x - x_i), j = 0, 1, \dots, n \quad (9)$$

where the value of the product is taken to be 1 when the limits make it meaningless. Thus, given the data (x_k, f_k) , $k = 0, 1, 2, \dots, n$, then

$$\phi_0(x) = \prod_{i=0}^{-1} (x - x_i) = 1$$

$$\phi_1(x) = \prod_{i=0}^0 (x - x_i) = x - x_0$$

$$\phi_2(x) = \prod_{i=0}^1 (x - x_i) = (x - x_0)(x - x_1)$$

$$\vdots$$

$$\phi_n(x) = \prod_{i=0}^{n-1} (x - x_i) = (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1})$$

Then, to find the parameters α_k we have that,

$$p_n(x_k) = \sum_{j=0}^n \alpha_j \phi_j(x_k) = f_k, \quad k = 0, 1, 2, \dots, n. \quad (10)$$

So we have the system

$$\begin{pmatrix} \phi_0(x_0) & \phi_1(x_0) & \cdots & \phi_n(x_0) \\ \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_n(x_1) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_n(x_n) \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_n \end{pmatrix}$$

that is

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{f} \quad (11)$$

where the matrix $\mathbf{A}$ is such that $a_{kj} = \phi_{j-1}(x_{k-1})$, $j, k = 0, 1, \dots, n+1$, and $a_{kj} = 0$, if $k < j$. This gives a lower diagonal matrix

$$\begin{pmatrix} \phi_0(x_0) & & & \\ \phi_0(x_1) & \phi_1(x_1) & & \\ \vdots & & \ddots & \\ \phi_0(x_n) & \phi_1(x_n) & \cdots & \phi_n(x_n) \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ \vdots \\ f_n \end{pmatrix}$$

which can be solved by forward substitution.

Example 2.2 Given the data

x	-2	0	1
$f(x)$	-27	-1	0

Find the Newton interpolating polynomial.

Solution: We are given $x_0 = -2$, $x_1 = 0$ and $x_2 = 1$. Then,

$$\phi_0(x) = \prod_{i=0}^{0-1} (x - x_i) = 1$$

$$\phi_1(x) = \prod_{i=0}^{1-1} (x - x_i) = (x - x_0)$$

$$\phi_2(x) = \prod_{i=0}^{2-1} (x - x_i) = (x - x_0)(x - x_1)$$

So we compute the coefficients from

$$\begin{aligned} \alpha_0 \phi_0(x_0) + \alpha_1 \phi_1(x_0) + \alpha_2 \phi_2(x_0) &= f_0 \\ \alpha_0 \phi_0(x_1) + \alpha_1 \phi_1(x_1) + \alpha_2 \phi_2(x_1) &= f_1 \\ \alpha_0 \phi_0(x_2) + \alpha_1 \phi_1(x_2) + \alpha_2 \phi_2(x_2) &= f_2 \end{aligned}$$

This gives the system

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 1 & 3 & 3 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \end{pmatrix} = \begin{pmatrix} -27 \\ -1 \\ 0 \end{pmatrix}.$$

Solving yields $\alpha_0 = -27$, $\alpha_1 = 13$, $\alpha_2 = -4$ and hence

$$p_2(x) = -27 + 13(x + 2) - 4x(x + 2) = -4x^2 + 5x - 1.$$

2.2.4 Divided differences

When we use the Lagrange basis, there is no constructive relationship between successive polynomials. That is, we cannot use say polynomial p_{n-1} to construct polynomial p_n . Each polynomial is constructed afresh and that makes the process of finding higher degree polynomials computationally expensive. Note that the Newton polynomials are given by

$$\begin{aligned} p_0(x) &= \alpha_0, \\ p_1(x) &= \alpha_0 + \alpha_1(x - x_0), \\ p_2(x) &= \alpha_0 + \alpha_1(x - x_0) + \alpha_2(x - x_0)(x - x_1), \\ &\vdots \\ p_n(x) &= \alpha_0 + \alpha_1(x - x_0) + \alpha_2(x - x_0)(x - x_1) + \cdots \\ &\quad + \alpha_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}). \end{aligned}$$

That is, the Newton polynomials can be constructed recursively. (*Determine the recursive formula*). Now, to find the coefficients $\alpha_0, \dots, \alpha_n$, we note that, for the polynomial $p_1(x)$, since $p_1(x_0) = f_0$ and $p_1(x_1) = f_1$, then

$$p_1(x_0) = \alpha_0 + \alpha_1(x_0 - x_0) = f_0 \implies \alpha_0 = f_0,$$

and, using this value of α_0 ,

$$p_1(x_1) = f_0 + \alpha_1(x_1 - x_0) = f_1 \implies \alpha_1 = \frac{f_1 - f_0}{x_1 - x_0}.$$

Now, evaluating $p_2(x)$ at x_2 and using the values of α_0 and α_1 obtained above gives us

$$\alpha_2 = \frac{\frac{f_2 - f_1}{x_2 - x_1} - \frac{f_1 - f_0}{x_1 - x_0}}{x_2 - x_0}.$$

The results for α_0 , α_1 and α_2 indicate to an alternative way of computing the coefficients of the Newton polynomial in what is known as *divided differences*.

Definition 2.2 Given a set of data points (x_i, f_i) , $i = 0, 1, 2, \dots, n$ the divided differences denoted by $f[\]$ are defined recursively by the formula;

$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}, \quad (12)$$

where the recursion begins with

$$f[x_k] = f_k, k = 0, 1, \dots, n.$$

More generally, we have

$$f[x_{k-j}, x_{k-j+1}, \dots, x_k] = \frac{f[x_{k-j+1}, x_{k-j+2}, \dots, x_k] - f[x_{k-j}, x_{k-j+1}, \dots, x_{k-1}]}{x_k - x_{k-j}}, \quad k > j. \quad (13)$$

The coefficients in the Newton interpolant are then given by

$$\alpha_j = f[x_0, x_1, \dots, x_j].$$

Example 2.3 Consider the previous example where the given data is

x	-2	0	1
$f(x)$	-27	-1	0

Then

$$\alpha_0 = f[x_0] = f_0 = -27,$$

$$\alpha_1 = f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{-1 - (-27)}{0 - (-2)} = \frac{26}{2} = 13,$$

$$\alpha_2 = f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}.$$

But

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{0 - -1}{1 - 0} = 1$$

So

$$\alpha_2 = \frac{1 - 13}{0 - -2} = \frac{-12}{3} = -4.$$

Therefore the Newton polynomial is

$$\begin{aligned} p_2(x) &= \alpha_0 + \alpha_1(x - x_0) + \alpha_2(x - x_0)(x - x_1) \\ &= -27 + 13(x - (-2)) - 4(x - (-2))(x - 0) \\ &= -4x^2 + 5x - 1 \end{aligned}$$

The computation of the divided differences can be organised in a table called a divided differences scheme as illustrated in Table 1.

x_k	0 th Order	1 st Order	2 nd Order
x_0	$f[x_0] = f_0$		
		$f[x_0, x_1]$	
x_1	$f[x_1] = f_1$		$f[x_0, x_1, x_2]$
		$f[x_1, x_2]$	
x_2	$f[x_2] = f_2$		

Table 1: A divided difference scheme for three nodes.

3 Finite Differences

Suppose $y = f(x)$ has values $y_0, y_1, \dots, y_n$ for the equally spaced values

$$x_0, x_0 + h, x_0 + 2h, \dots, x_0 + nh, \quad (14)$$

for some $h > 0$. That is

$$x_i = x_0 + ih, \text{ and } y_i = f(x_i) \text{ for } i = 0, 1, \dots, n. \quad (15)$$

To determine other values of say y , $\frac{dy}{dx}$ for some intermediate value of x , we base on the principle of finite differences.

3.1 Forward Differences

Given $y_i = f(x_i)$ is as defined in (15), the forward difference at y_i is defined as

$$\Delta y_i = y_{i+1} - y_i, \text{ for } i = 0, 1, \dots, n-1. \quad (16)$$

Therefore

$$\Delta y_0 = y_1 - y_0, \Delta y_1 = y_2 - y_1, \dots, \Delta y_{n-1} = y_n - y_{n-1},$$

are the forward differences of $f(x)$ given the nodes (14). Generally these are called the *first order forward differences* or simply the *first forward differences* of f . Here Δ is called the *forward difference operator*. The differences of the first forward differences are called the *second order forward differences* or simply the *second differences* of f . That is

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0, \Delta^2 y_1 = \Delta y_2 - \Delta y_1, \dots, \Delta^2 y_{n-1} = \Delta y_n - \Delta y_{n-1}.$$

Generally the n -th order forward differences are defined as

$$\Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i. \quad (17)$$

Higher order differences can also be written in terms of the y_i s. For instance,

$$\begin{aligned} \Delta^2 y_i &= \Delta y_{i+1} - \Delta y_i \\ &= (y_{i+2} - y_{i+1}) - (y_{i+1} - y_i) \\ &= y_{i+2} - 2y_{i+1} + y_i. \end{aligned}$$

The computation of the forward differences can be arranged in a table called the *forward differences table*. For instances for $n = 4$, we have the differences in Table 2

x_i	y_i	1 st Order	2 nd Order	3 rd Order	4 th Order
x_0	y_0				
		Δy_0			
x_1	y_1		$\Delta^2 y_0$		
		Δy_1		$\Delta^3 y_0$	
x_2	y_2		$\Delta^2 y_1$		$\Delta^4 y_0$
		Δy_2		$\Delta^3 y_1$	
x_3	y_3		$\Delta^2 y_2$		
		Δy_3			
x_4	y_4				

Table 2: A forward differences table for (x_i, y_i) , $i = 0, 1, 2, 3, 4$.

3.2 Backward Differences

Given the data (15), the *first order backward difference* or simply the *first backward difference* at y_i , denoted ∇y_i , is defined as

$$\nabla y_i = y_i - y_{i-1}, \text{ for } i = 1, 2, \dots, n. \quad (18)$$

Here, ∇ is called the *backward difference operator*. Generally, the n -th order backward difference is defined as

$$\nabla^n y_i = \nabla^{n-1} y_i - \nabla^{n-1} y_{i-1}. \quad (19)$$

Higher order backward differences can also be written in terms of y_i s. For instance, the second backward difference at y_i is

$$\begin{aligned} \nabla^2 y_i &= \nabla y_i - \nabla y_{i-1} \\ &= (y_i - y_{i-1}) - (y_{i-1} - y_{i-2}) \\ &= y_i - 2y_{i-1} + y_{i-2}. \end{aligned}$$

The process of computing backward differences can be summarised in a table called a backward differences table as shown in Table 3.

3.3 Central Differences

Given data (x_i, y_i) as defined in (15), the *central differences*, denoted δy_i are defined as

$$\delta y_i = y_{i+\frac{1}{2}} - y_{i-\frac{1}{2}}. \quad (20)$$

Similar to the definition of y_i , $y_{i+\frac{1}{2}}$ is given by

$$y_{i+\frac{1}{2}} = f(x_{i+\frac{1}{2}}), \text{ where } x_{i+\frac{1}{2}} = x_i + \frac{1}{2}h.$$

x_i	y_i	1 st Order	2 nd Order	3 rd Order	4 th Order
x_0	y_0				
		∇y_1			
x_1	y_1		$\nabla^2 y_2$		
		∇y_2		$\nabla^3 y_3$	
x_2	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$
		∇y_3		$\nabla^3 y_4$	
x_3	y_3		$\nabla^2 y_4$		
		∇y_4			
x_4	y_4				

Table 3: A table of backward differences for (x_i, y_i) , $i = 0, 1, 2, 3, 4$.

The operator δ is called the *central difference operator*. It can be applied at any point. For instance

$$\delta y_{\frac{1}{2}} = y_{\frac{1}{2}+\frac{1}{2}} - y_{\frac{1}{2}-\frac{1}{2}} = y_1 - y_0,$$

$$\delta y_{\frac{3}{2}} = y_{\frac{3}{2}+\frac{1}{2}} - y_{\frac{3}{2}-\frac{1}{2}} = y_2 - y_1,$$

$$\delta y_{\frac{5}{2}} = y_{\frac{5}{2}+\frac{1}{2}} - y_{\frac{5}{2}-\frac{1}{2}} = y_3 - y_2,$$

$$\delta y_3 = y_{3+\frac{1}{2}} - y_{3-\frac{1}{2}} = y_{\frac{7}{2}} - y_{\frac{5}{2}}.$$

All these are first order central differences. In general, the n -th order central difference is defined as

$$\delta^n y_i = \delta^{n-1} y_{i+\frac{1}{2}} - \delta^{n-1} y_{i-\frac{1}{2}}. \quad (21)$$

For instance,

$$\delta^2 y_2 = \delta y_{2+\frac{1}{2}} - \delta y_{2-\frac{1}{2}} = \delta y_{\frac{5}{2}} - \delta y_{\frac{3}{2}}.$$

The computation of central differences can also be arranged in a table called a table of central differences as shown in Table 4 for the case of $n = 4$.

Example 3.1 Given the data

x	1	2	3	4	5
y	2	5	10	20	30

x_i	y_i	1 st Order	2 nd order	3 rd Order	4 th Order
x_0	y_0	$\delta y_{\frac{1}{2}}$			
x_1	y_1	$\delta y_{\frac{3}{2}}$	$\delta^2 y_1$	$\delta^3 y_{\frac{3}{2}}$	
x_2	y_2	$\delta y_{\frac{5}{2}}$	$\delta^2 y_2$	$\delta^3 y_{\frac{5}{2}}$	$\delta^4 y_2$
x_3	y_3	$\delta y_{\frac{7}{2}}$	$\delta^2 y_3$		
x_4	y_4				

Table 4: A table of central differences for (x_i, y_i) , $i = 0, 1, 2, 3, 4$.

(a) Construct a forward differences table for the data.

(b) Give the following from the table:

(i) $\Delta^4 y(1)$

(ii) $\Delta^3 y(1)$

(iii) $\delta y_{\frac{1}{2}}$

(iv) ∇y_2

(v) Δy_2

Solution:

(a) Table 5 is the table of forward differences.

(b) In this example we see that we can read different differences from the same table:

(i) $\Delta^4 y(1) = \Delta^4 y_0 = -8$

(ii) $\Delta^3 y(1) = \Delta^3 y_0 = 3$

(iii) $\delta y_{\frac{1}{2}} = y_{\frac{1}{2}+\frac{1}{2}} - y_{\frac{1}{2}-\frac{1}{2}} = \Delta y_0 = 3$

(iv) $\nabla y_2 = y_2 - y_1 = \Delta y_1 = 5$

(v) $\Delta y_2 = y_3 - y_2 = 10$

Example 3.2 Given $y(x) = x^2 + 1$, use a step size of $h = 0.2$ to construct a table of backward differences for y on $[1, 2]$.

x_i	y_i	1 st Order	2 nd Order	3 rd Order	4 th Order
1	2				
		3			
2	5		2		
		5		3	
3	10		5		-8
		10		-5	
4	20		0		
		10			
5	30				

Table 5: A forward differences table for Example 3.1.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
1.0	2					
		0.44				
1.2	2.44		0.08			
		0.52		0		
1.4	2.96		0.08		0	
		0.60		0		0
1.6	3.56		0.08		0	
		0.68		0		
1.8	4.24		0.08			
		0.72				
2.0	5					

Table 6: A table of backward differences for Example 3.2.

Solution: Now, $x_i = x_0 + ih, i = 0, 1, \dots, n$. In this case, $x_0 = 1$. So we have the nodes 1, 1.2, 1.4, 1.6, 1.8, 2.0. The differences are in Table 6. Higher order differences in this example become zero. This phenomenon is what is known as convergence of the differences.

Definition 3.1 Differences are said to converge if $|\Delta^n y| \rightarrow 0$ as $n \rightarrow \infty$. Otherwise they are said to be divergent.

3.4 The shift operator

The *shift (increment) operator* E is defined as

$$Ey_i = y_{i+1}, \text{ for } i = 0, 1, 2, \dots, n-1. \quad (22)$$

Higher order shifts are defined recursively as

$$E^n y_i = E(E^{n-1} y_i). \quad (23)$$

For instance

$$E^2 y_i = E(Ey_i) = Ey_{i+1} = y_{i+2}.$$

Note that y_{i+2} means $y(x_i + 2h)$. Generally,

$$E^n y_i = y_{i+n}. \quad (24)$$

The *inverse shift operator* E^{-1} is defined by

$$E^{-1} y_i = y_{i-1}. \quad (25)$$

3.5 The averaging operator

The averaging operator μ is defined as

$$\mu = \frac{E^{\frac{1}{2}} + E^{-\frac{1}{2}}}{2}. \quad (26)$$

Note here also that $E^{-\frac{1}{2}} y_i = y(x_i - \frac{1}{2}h)$. That is

$$\mu y_i = \left(\frac{E^{\frac{1}{2}} + E^{-\frac{1}{2}}}{2} \right) y_i = \frac{E^{\frac{1}{2}} y_i + E^{-\frac{1}{2}} y_i}{2} = \frac{1}{2} (y_{i+\frac{1}{2}} + y_{i-\frac{1}{2}}).$$

3.6 Relations between the operators

We can establish several relations between the above operators. Example 3.3 gives some of them.

Example 3.3 Show that

- (a) $\Delta y_i = \nabla y_{i+1}$
- (b) $\Delta = E - 1$ or $E = \Delta + 1$
- (c) $\nabla = 1 - E^{-1}$
- (d) $E\nabla = \nabla E = \Delta$
- (e) $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$
- (f) $\Delta = E^{\frac{1}{2}}\delta$

Solution:

- (a) $\Delta y_i = y_{i+1} - y_i = \nabla y_{i+1}$
- (b) $\Delta y_i = y_{i+1} - y_i = Ey_i - y_i = (E - 1)y_i. \implies \Delta = E - 1$ or $E = \Delta + 1$.
- (c) $\nabla y_i = y_i - y_{i-1} = y_i - E^{-1}y_i = (1 - E^{-1})y_i. \implies \nabla = 1 - E^{-1}$ or $E^{-1} = \nabla + 1$, where 1 is the identity operator.
- (d) Starting with $E\nabla y_i$ we have

$$E\nabla y_i = E(y_i - y_{i-1}) = Ey_i - Ey_{i-1} = y_{i+1} - y_i = \Delta y_i.$$

Also

$$\nabla Ey_i = \nabla y_{i+1} = y_{i+1} - y_i = \Delta y_i.$$

Combining the two results we have that $E\nabla = \nabla E = \Delta$.

$$(e) \delta y_i = y_{i+\frac{1}{2}} - y_{i-\frac{1}{2}} = E^{\frac{1}{2}}y_i - E^{-\frac{1}{2}}y_i = (E^{\frac{1}{2}} - E^{-\frac{1}{2}})y_i \implies \delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}.$$

$$(f) \Delta y_i = y_{i+1} - y_i = Ey_i - y_i = E^{\frac{1}{2}}E^{\frac{1}{2}}y_i - E^{-\frac{1}{2}}E^{\frac{1}{2}}y_i = E^{\frac{1}{2}}(E^{\frac{1}{2}} - E^{-\frac{1}{2}})y_i = E^{\frac{1}{2}}\delta y_i.$$

Hence $\Delta = E^{\frac{1}{2}}\delta$.

Exercise 3.6-1: Show that

$$f[x_0, x_1, \dots, x_k] = \frac{\Delta^k f_0}{k!h^k}.$$

Theorem 3.1 (Fundamental Theorem of Differences calculus) *If $f(x)$ is a polynomial of n^{th} degree in x , then the n^{th} difference of $f(x)$ is a constant and*

$$\Delta^{n+1}f(x) = 0.$$

Proof

Consider $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$, $a_i \in \mathbb{R} \forall i$ and $n \in \mathbb{Z}^+$. By definition

$$\Delta f(x) = f(x+h) - f(x)$$

$$= [a_0 + a_1(x+h) + a_2(x+h)^2 + \cdots + a_n(x+h)^n] - [a_0 + a_1x + a_2x^2 + \cdots + a_nx^n]$$

$$= a_1[(x+h) - (x)] + a_2[(x+h)^2 - x^2] + \cdots + a_n[(x+h)^n - x^n]$$

Recall

$$(x+h)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} h^r.$$

Using this in the above difference and simplifying we obtain

$$\Delta f(x) = b_0 + b_1x + b_2x^2 + \cdots + b_{n-1}x^{n-1}$$

where b_i , $i = 0, 1, 2, \dots, n$ are constants depending on h and a_i , $i = 0, 1, 2, \dots, n$, and

$$b_{n-1} = a_n \binom{n}{1} h.$$

Thus $\Delta f(x)$ is a polynomial of degree $n-1$. Like wise, let us set

$$g(x) = b_0 + b_1x + b_2x^2 + \cdots + b_{n-1}x^{n-1}.$$

Then

$$\begin{aligned} \Delta^2 f(x) &= \Delta[f(x+h) - f(x)] \\ &= \Delta g(x) \\ &= g(x+h) - g(x) \end{aligned}$$

will give a polynomial of degree $n-2$

$$h(x) = c_0 + c_1x + c_2x^2 + \cdots + c_{n-2}x^{n-2}.$$

Exercise 3.6-2: Deduce c_{n-2} in terms of b_{n-1} , n and h .

Proceeding in a similar manner, $\Delta^n f(x)$ will give a polynomial of degree $(n - n)$ which is a constant, say K .

Exercise 3.6-3: Deduce K in terms of n , a_n and h .

Then $\Delta^{n+1} f(x) = \Delta K = K - K = 0$.

Exercise 3.6-4:

Prove the following relations

$$(a) \quad E = (1 - \nabla)^{-1}$$

$$(b) \quad \delta = \Delta(1 + \Delta)^{-1}$$

$$(c) \quad D = h^{-1} \ln E$$

3.7 Newton's Forward and Backward difference interpolation formulas

Let $x_0, x_1, \dots, x_n$ be equally spaced nodes as defined in (15). Let also $f_n(x)$ be a polynomial of degree n that interpolates (x_i, y_i) , $i = 0, 1, \dots, n$. Then using the Newton basis we have that

$$f_n(x) = A_0 + A_1(x - x_0) + A_2(x - x_0)(x - x_1) + \dots + A_n(x - x_0)(x - x_1)(x - x_2) \dots (x - x_{n-1})$$

Since $f_n(x)$ interpolates (x_i, y_i) , substituting $x = x_0$ we have

$$f_n(x_0) = y_0 = A_0.$$

Substituting $x = x_1$ we have

$$f_n(x_1) = y_1 = A_0 + A_1(x_1 - x_0) \implies A_1 = \frac{y_1 - A_0}{x_1 - x_0} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y_0}{h}.$$

Substituting $x = x_2$ we have

$$f_n(x_2) = y_2 = A_0 + A_1(x_2 - x_0) + A_2(x_2 - x_0)(x_2 - x_1).$$

Then

$$y_2 = y_0 + \frac{\Delta y_0}{h}(2h) + A_2(2h)h = y_0 + 2\Delta y_0 + 2A_2h^2,$$

which gives

$$A_2 = \frac{y_2 - y_0 - 2\Delta y_0}{2h^2} = \frac{y_2 - y_0 - 2(y_1 - y_0)}{2h^2} = \frac{y_2 - 2y_1 + y_0}{2h^2} = \frac{\Delta^2 y_0}{2h^2} = \frac{\Delta^2 y_0}{2!h^2}.$$

Proceeding in a similar way gives

$$A_3 = \frac{\Delta^3 y_0}{3!h^3}, \quad A_4 = \frac{\Delta^4 y_0}{4!h^4}, \quad \dots, \quad A_n = \frac{\Delta^n y_0}{n!h^n}.$$

Thus the Newton interpolant is given by

$$\begin{aligned} f_n(x) = y_0 + \frac{\Delta y_0}{h}(x - x_0) + \frac{\Delta^2 y_0}{2!h^2}(x - x_0)(x - x_1) + \frac{\Delta^3 y_0}{3!h^3}(x - x_0)(x - x_1)(x - x_2) \\ + \dots + \frac{\Delta^n y_0}{n!h^n}(x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1}) \quad (27) \end{aligned}$$

Set $x = x_0 + ph$, that is, $x - x_0 = ph$ for some parameter p . Then

$$\begin{aligned} x - x_0 &= ph, \\ x - x_1 &= x - (x_0 + h) = x - x_0 - h = ph - h = (p - 1)h, \\ x - x_2 &= x - (x_0 + 2h) = x - x_0 - 2h = ph - 2h = (p - 2)h. \end{aligned}$$

Proceeding in a similar manner we get

$$x - x_3 = (p - 3)h, \quad x - x_4 = (p - 4)h, \quad \dots, \quad x - x_{n-1} = (p - (n - 1))h.$$

We can then write the factors $x - x_i$ in terms of p and h as

$$\begin{aligned} x - x_0 &= ph, \\ (x - x_0)(x - x_1) &= p(p - 1)h^2, \\ (x - x_0)(x - x_1)(x - x_2) &= p(p - 1)(p - 2)h^3, \\ &\vdots \\ (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1}) &= p(p - 1)(p - 2) \cdots (p - (n - 1))h^n. \end{aligned}$$

Also

$$\begin{aligned}
 \frac{\Delta y_0}{h}(x - x_0) &= p\Delta y_0, \\
 \frac{\Delta^2 y_0}{2!h^2}(x - x_0)(x - x_1) &= \frac{p(p-1)}{2!}\Delta^2 y_0, \\
 \frac{\Delta^3 y_0}{3!h^3}(x - x_0)(x - x_1)(x - x_2) &= \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0, \\
 &\vdots \\
 \frac{\Delta^n y_0}{n!h^n}(x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{n-1}) &= \frac{p(p-1)(p-2) \cdots (p-(n-1))}{n!}\Delta^n y_0.
 \end{aligned}$$

Hence

$$\begin{aligned}
 f_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 \\
 + \cdots + \frac{p(p-1)(p-2) \cdots (p-(n-1))}{n!}\Delta^n y_0, \quad (28)
 \end{aligned}$$

where $p = (x - x_0)/h$ is Newton's *forward difference interpolation formula*.

Exercise 3.7-1: Show that the Newton *backward difference interpolation formula* is

$$\begin{aligned}
 f_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n \\
 + \cdots + \frac{p(p+1)(p+2) \cdots (p+(n-1))}{n!}\nabla^n y_n, \quad (29)
 \end{aligned}$$

where $p = \frac{x - x_n}{h}$.

4 Numerical Differentiation

We usually consider numerical differentiation when:

- (i) we want to compute the derivatives of a function which is known only at a discrete set of grid points,
- (ii) we want to approximate derivatives in ordinary and partial differential equations to reduce the equations to an easy-to-solve form of equation,
- (iii) we want to approximate derivatives of an otherwise difficult-to-differentiate function.

4.1 Finite differences approximations

Finite difference formulas are very useful for approximating derivatives of a smooth function that:

- (a) is defined on a discrete set of grid points but is otherwise known analytically,
- (b) can be evaluated accurately for any given argument,
- (c) is defined implicitly by a differential equation.

Consider $f : \mathbb{R} \rightarrow \mathbb{R}$ a smooth function. Let $h > 0$ be a small distance usually called a step size. We wish to approximate the derivatives of f at $x \in \mathbb{R}$ for any step size h . Consider the Taylor series expansion of $f(x + h)$,

$$f(x + h) = f(x) + f'(x)h + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \cdots + f^{(n)}(x)\frac{h^n}{n!} + \cdots \quad (30a)$$

Or

$$f(x - h) = f(x) - f'(x)h + f''(x)\frac{h^2}{2!} - f'''(x)\frac{h^3}{3!} + \cdots + f^{(n)}(x)\frac{h^n}{n!} + \cdots \quad (30b)$$

Rearranging (30a) we obtain

$$f'(x) = \frac{f(x + h) - f(x)}{h} - f''(x)\frac{h}{2!} - f'''(x)\frac{h^2}{3!} - \cdots - f^{(n)}(x)\frac{h^{n-1}}{n!} - \cdots \quad (31)$$

That is

$$f'(x) = \frac{f(x+h) - f(x)}{h} + \mathcal{O}(h). \quad (32)$$

From (32) we take

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad (33)$$

and the error we make, called *truncation error* is $\mathcal{O}(h)$ since the truncated term,

$$-f''(x)\frac{h}{2!} - f'''(x)\frac{h^2}{3!} - \dots - f^{(n)}(x)\frac{h^{n-1}}{n!} - \dots,$$

is dominated by the term $-f''(x)\frac{h}{2!}$ which is $\mathcal{O}(h)$. Actually, there exists an $\eta \in [x, x+h]$ such that the error E is

$$E = -f''(\eta)\frac{h}{2}. \quad (34)$$

The formula (33) is called the *forward difference formula* for f' at x . It is an *order one* (or *first order*) method meaning that the error is $\mathcal{O}(h)$. Geometrically, the formula (33) can be illustrated as shown in Figure 1: the slope of the chord AB is used to approximate the slope of the tangent line at x .

Exercise 4.1-1: In Figure 1, we have that

$$f'(x) \approx \text{gradient/slope of segment AB} = \frac{f(x+h) - f(x)}{h}$$

and as $h \rightarrow 0$, $AB \rightarrow$ tangent at A. How fast does $AB \rightarrow$ tangent at A. In other words, how fast does the error committed in (33) go to zero?

Now using (30b) we have

$$f'(x) = \frac{f(x) - f(x-h)}{h} + f''(x)\frac{h}{2!} - f'''(x)\frac{h^2}{3!} + \dots. \quad (35)$$

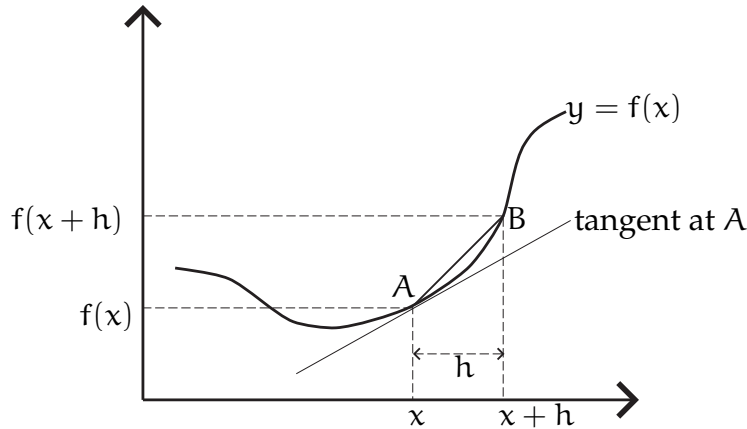


Figure 1: Geometrical illustration of the forward difference formula.

This gives

$$f'(x) \approx \frac{f(x) - f(x - h)}{h}. \quad (36)$$

Formula (36) is called the *backward difference formula* for $f'(x)$. It is an *order one* (or *first order*) method since the truncation error is $\mathcal{O}(h)$. To be precise, there exists a $\beta \in [x - h, x]$ such that the truncation error of (36) is

$$E = f''(\beta) \frac{h}{2!}. \quad (37)$$

Geometrically, the formula (36) is illustrated in Figure 2.

Exercise 4.1-2: In Figure 2, we have that

$$f'(x) \approx \text{gradient/slope of segment CA} = \frac{f(x) - f(x - h)}{h}$$

and as $h \rightarrow 0$, $CA \rightarrow \text{tangent at A}$. How fast does $CA \rightarrow \text{tangent at A}$. In other-wards, how fast does the error committed in (36) go to zero?

Now, if we subtract (30b) from (30a) we obtain

$$f(x + h) - f(x - h) = 2f'(x)h + 2f'''(x)\frac{h^3}{3!} + \dots. \quad (38)$$

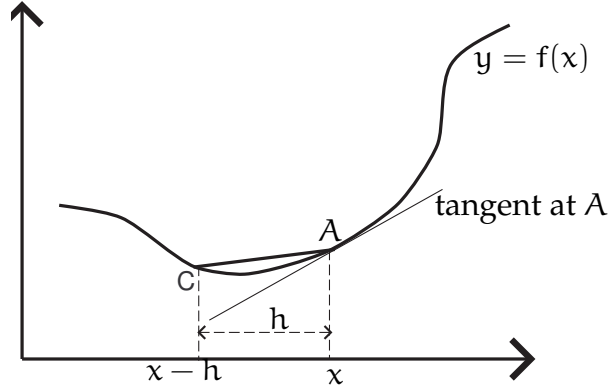


Figure 2: Geometrical illustration of the backward differences formula.

Rearranging this gives

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} - f'''(x) \frac{h^2}{3!} - \dots, \quad (39)$$

so that

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}. \quad (40)$$

The formula (40) is called the *central difference formula* for $f'(x)$. It is an *order two* (*second order*) formula. That is, the truncation error is $\mathcal{O}(h^2)$. To be precise, there is a $\gamma \in [x-h, x+h]$ such that

$$E = -f'''(\gamma) \frac{h^2}{6}. \quad (41)$$

We say that it has *Quadratic convergence*, whereas (33) and (36) are said to have linear convergence. Geometrically, the central difference formula is illustrated in Figure 3.

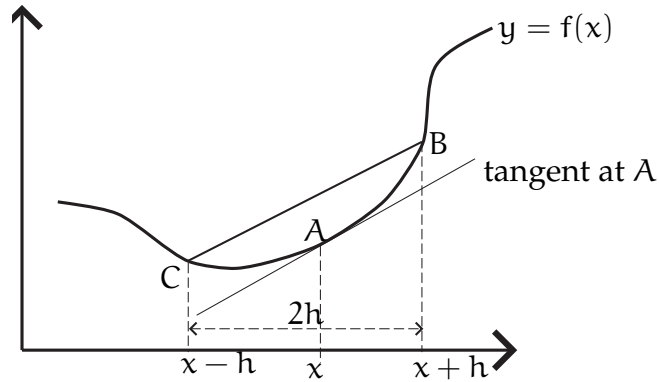


Figure 3: Central differences.

Exercise 4.1-3: In Figure 3 we have that

$$f'(x) \approx \text{gradient of segment CB} = \frac{f(x+h) - f(x-h)}{2h},$$

and, like in the backward and forward formulas,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h} = \text{slope of tangent at A}.$$

How fast does CB tend to tangent at A. In otherwords, how fast does the error committed in (40) go to zero?

Example 4.1 Given $f(x) = x^2$. Using a step size of (a) $h = 0.1$ and (b) $h = 0.01$, compute $f'(2)$ using the

- (i) Forward difference formula,
- (ii) Backward difference formula,
- (iii) Central difference formula.

(iv) Compute and explain the errors in each of your estimates.

Solution:

- (a) (i) Using forward differences

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \implies f'(2) \approx \frac{(2+0.1)^2 - 2^2}{0.1} = \frac{0.41}{0.1} = 4.1.$$

(ii) Using backward differences

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}, \Rightarrow f'(2) \approx \frac{2^2 - (2-0.1)^2}{0.1} = \frac{0.39}{0.1} = 3.9.$$

(iii) Using central differences

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \Rightarrow f'(2) \approx \frac{(2+0.1)^2 - (2-0.1)^2}{2(0.1)} = \frac{0.8}{0.2} = 4$$

(iv) In this case we can compute the errors using the exact value: From calculus, $f'(x) = (x^2)' = 2x$. Hence, $f'(2) = 2(2) = 4$. Thus the errors are

$$\text{In (i)} \quad E = |\text{exact} - \text{approximate}| = |4 - 4.1| = 0.1.$$

$$\text{In (ii)} \quad E = |\text{exact} - \text{approximate}| = |4 - 3.9| = 0.1.$$

$$\text{In (iii)} \quad E = |\text{exact} - \text{approximate}| = |4 - 4| = 0.$$

In (i) and (ii), we use first order formulas to estimate the derivative of a quadratic polynomial, we can expect nonzero error. In (iii), we use a second order method to estimate the derivative of a second order polynomial. It is expected to be exact. Using (41), we can see that indeed the error will be zero due to the term $f'''(\gamma)$.

(b) (i) Using forward differences

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \Rightarrow f'(2) \approx \frac{(2+0.01)^2 - 2^2}{0.01} = 4.01.$$

(ii) Using backward differences

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}, \Rightarrow f'(2) \approx \frac{2^2 - (2-0.01)^2}{0.01} = 3.99.$$

(iii) Using central differences

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \Rightarrow f'(2) \approx \frac{(2+0.01)^2 - (2-0.01)^2}{2(0.01)} = 4.$$

(iv) Again using the exact value, the errors are

$$\text{In (i) } E = |\text{exact} - \text{approximate}| = |4 - 4.01| = 0.01.$$

$$\text{In (ii) } E = |\text{exact} - \text{approximate}| = |4 - 3.99| = 0.01.$$

$$\text{In (iii) } E = |\text{exact} - \text{approximate}| = |4 - 4| = 0.$$

In (i) and (ii), we see that when h is reduced by a factor 10, the error also goes down by a factor 10, that is, by the same factor as h . This is the case for they are order 1 methods. In (iii), the error is again zero for same reasons as above. What would be the reduction factor in the error if the error was not zero?

Exercise 4.1-4: Use formulas (34), (37) and (41) to estimate the errors in Example 4.1.

If we now add the expansions (30a) and (30b) we get

$$f(x+h) + f(x-h) = 2f(x) + 2f''(x)\frac{h^2}{2!} + 2f^{(4)}(x)\frac{h^4}{4!} + 2f^{(6)}(x)\frac{h^6}{6!} + \dots$$

Rearranging gives

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - f^{(4)}(x)\frac{h^2}{12} - 2f^{(6)}(x)\frac{h^4}{6!} - \dots$$

We can then make the approximation

$$f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}. \quad (42)$$

What we have in (42) is called the *central difference formula* for $f''(x)$. The truncation

error we make is dominated by the term $f^{(4)}(x)\frac{h^2}{12}$ which is $\mathcal{O}(h^2)$. Therefore it is a *second order (order 2) method*, that is, it has *quadratic convergence*. To be precise, there exists $\alpha \in [x-h, x+h]$ so that the error is

$$E = -f^{(4)}(\alpha)\frac{h^2}{12}. \quad (43)$$

4.2 Differentiation using the finite difference polynomial interpolant

Recall the Newton forward and backward difference interpolation formulas (28) and (29) of Section 3, that is,

$$f_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots + \frac{p(p-1)(p-2)\dots(p-(n-1))}{n!}\Delta^n y_0, \quad (44)$$

where $p = (x - x_0)/h$, and

$$f_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots + \frac{p(p+1)(p+2)\dots(p+(n-1))}{n!}\nabla^n y_n, \quad (45)$$

where $p = \frac{x - x_n}{h}$.

Consider the forward difference interpolant (44). Using the expression for p and the chain rule we have

$$\frac{df}{dx} = \frac{df}{dp} \cdot \frac{dp}{dx} = \frac{1}{h} \cdot \frac{df}{dp}. \quad (46)$$

Thus

$$\frac{df}{dx} = \frac{1}{h} \left(\Delta y_0 + \frac{2p-1}{2!}\Delta^2 y_0 + \frac{3p^2-6p+2}{3!}\Delta^3 y_0 + \dots + \left[\frac{p(p-1)(p-2)\dots(p-(n-1))}{n!} \right]' \Delta^n y_0 \right). \quad (47)$$

This equation can be used to find $\frac{df}{dx}$ at a general non-tabular point x or at a tabular point x_0 . At a tabular point $x = x_0$, $p = 0$ so that we get

$$\frac{df}{dx} = \frac{1}{h} \left(\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right), \quad (48)$$

which can be used to estimate df/dx at a tabular point x_i , $i = 0, 1, 2, \dots, n$. That is,

$$\left. \frac{df}{dx} \right|_{x_i} = \frac{1}{h} \left(\Delta y_i - \frac{1}{2} \Delta^2 y_i + \frac{1}{3} \Delta^3 y_i - \frac{1}{4} \Delta^4 y_i + \dots \right), \quad i = 0, 1, \dots, n. \quad (49)$$

In a similar way, formulas for higher order derivatives can be derived from (44). For instance, from the expression for df/dx ,

$$\frac{d^2f}{dx^2} = \frac{d}{dx} \left(\frac{1}{h} \frac{df}{dp} \right) = \frac{d}{dp} \left(\frac{1}{h} \frac{df}{dp} \right) \frac{dp}{dx} = \frac{1}{h^2} \frac{d^2f}{dp^2}. \quad (50)$$

Substituting d^2f/dp^2 gives

$$\frac{d^2f}{dx^2} = \frac{1}{h^2} \cdot \left(\Delta^2 y_0 + \frac{6p-6}{3!} \Delta^3 y_0 + \dots + \left[\frac{p(p-1)(p-2) \dots (p-(n-1))}{n!} \right]'' \Delta^n y_0 \right). \quad (51)$$

which can be used to estimate d^2f/dx^2 at a point p .

Example 4.2 Given

x	1	2	3	4
y	4	9	16	25

Construct a forward difference scheme for the data and find $\frac{dy}{dx}$ at

(i) $x = 1$

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1	4			
		5		
2	9		2	
		7		0
3	16		2	
		9		
4	25			

Table 7: Table of forward differences for Example 4.2.

(ii) $x = 2$ (iii) $x = \frac{3}{2}$ **Solution:** The table of forward differences is Table 7.(i) Since $x = 1$ is a tabular point, we use the formula (48). That is

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{h} \left(\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \cdots \right) \\ &= \frac{1}{1} \left(5 - \frac{1}{2}(2) + \frac{1}{3}(0) \right) = 5 - 1 = 4.\end{aligned}$$

(ii) Again, $x = 2$ is a tabular point, so we use (48) starting at x_1 . That is

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{h} \left(\Delta y_1 - \frac{1}{2} \Delta^2 y_1 + \frac{1}{3} \Delta^3 y_1 + \cdots \right) \\ &= \frac{1}{1} \left(7 - \frac{1}{2} \cdot 2 \right) = 7 - 1 = 6\end{aligned}$$

(iii) This is not a tabular point so we find p and use the general formula. At $x =$

$\frac{3}{2}$, $p = \frac{x - x_0}{h} = \frac{\frac{3}{2} - 1}{1} = \frac{1}{2}$. Then

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{h} \left(\Delta y_0 + \frac{2p-1}{2!} \Delta^2 y_0 + \frac{3p^2-6p+2}{3!} \Delta^3 y_0 + \dots \right) \\ &= \frac{1}{1} \left(5 + \frac{2(\frac{1}{2})-1}{2!} (2) + \frac{3(\frac{1}{2})^2-6(\frac{1}{2})+2}{3} (0) \right) \\ &= 1(5 + 0 + 0) = 5. \end{aligned}$$

5 Numerical Integration

We consider numerical integration when we need to evaluate the intergral

$$I = \int_a^b f(x)dx \quad (52)$$

but

- (i) either the integrand $f(x)$ is too complicated to obtain the anti derivatives, or
- (ii) $f(x)$ is only known at some discrete points.

Let the interval $[a, b]$ be divided into n equal sub intervals

$$[x_0, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, x_n]$$

such that

$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_n = b \quad (53)$$

is a partition of $[a, b]$ and

$$x_i = x_0 + ih, i = 0, 1, 2, \dots, n.$$

Then

$$\int_a^b f(x)dx = \int_{x_0}^{x_n} f(x)dx$$

Let $y_i = f(x_i)$, approximating $f(x)$ by Newton's forward difference interpolant using $x_0, x_1, \dots, x_n$, we have

$$I \approx \int_{x_0}^{x_n} \left(y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)\Delta^3}{3!}y_0 + \dots + \frac{p(p-1)(p-2)\dots(p-(n-1))}{n!}\Delta^n y_0 \right) dx. \quad (54)$$

Now $x = x_0 + ph$, so that $dx = hdp$ and when $x = x_0$, $p = 0$, when $x = x_n$, $p = n$. Therefore

$$I \approx h \int_0^n \left(y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots + \frac{p(p-1)(p-2)\dots(p-(n-1))}{n!}\Delta^n y_0 \right) dp \quad (55)$$

Integrating (55) term by term gives

$$I \approx h \left(y_0 n + \frac{n^2}{2}\Delta y_0 + \frac{(n^2)(2n-3)}{12}\Delta^2 y_0 + \frac{(n^2)(n-2)^2}{24}\Delta^3 y_0 + \dots \right), \quad (56)$$

which is a general numerical integration formula for a given number of subintervals n .

5.1 Trapezoidal Rule

Consider integration using only one subinterval $[x_0, x_1]$, that is, $n = 1$, then the formula (56) gives

$$I \approx h \left(y_0 + \frac{1}{2}\Delta y_0 \right) = h \left(y_0 + \frac{1}{2}(y_1 - y_0) \right).$$

That is

$$\int_{x_0}^{x_1} f(x)dx = \int_{x_0}^{x_1} f(x)dx \approx \frac{h}{2}(y_0 + y_1) \quad (57)$$

Formula (57) is the *Trapezoidal rule* for $n = 1$, that is, one subinterval. However, in the composite trapezoidal rule, more than one subintervals are used and then (57) is applied on each subinterval. The integral on $[x_0, x_n] = [a, b]$ is then the sum of the integrals on the subintervals. That is, if we use $n > 1$ integrals, we have

$$\int_{x_0}^{x_1} f(x)dx \approx \frac{h}{2}(y_0 + y_1), \int_{x_1}^{x_2} f(x)dx \approx \frac{h}{2}(y_1 + y_2), \dots, \int_{x_{n-1}}^{x_n} f(x)dx \approx \frac{h}{2}(y_{n-1} + y_n). \quad (58)$$

The combined result is

$$I = \int_{x_0}^{x_n} f(x)dx \approx \sum_{i=1}^n \frac{h}{2}(y_{i-1} + y_i) = \frac{h}{2}(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n).$$

Therefore we have

$$\int_{x_0}^{x_n} f(x)dx \approx \frac{h}{2}(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n) \quad (59)$$

which is the *composite trapezoidal rule*, generally just known as the *trapezoidal rule* for estimating the integral (52) using n uniform subintervals defined by the partition (53). Figure 4 shows the geometrical illustration of the trapezoidal rule.

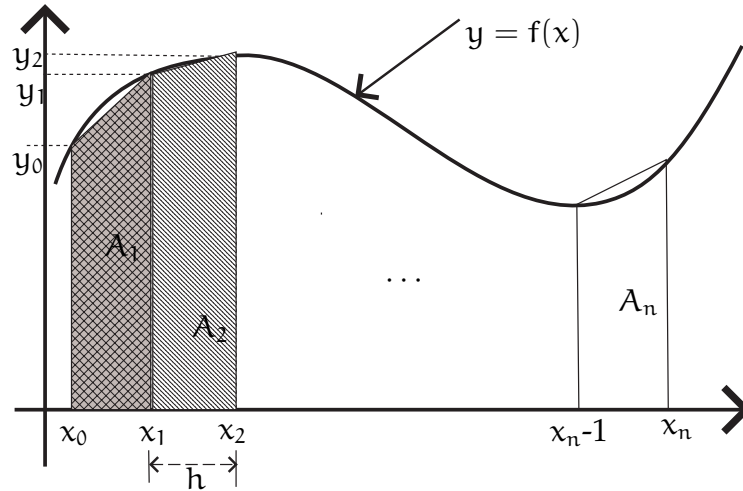


Figure 4: Illustration of integration using the trapezoidal rule.

The area under $f(x)$ on each subinterval $[x_{i-1}, x_i]$, $i = 0, 1, 2, \dots, n$, is approximated by the area of the trapezium. That is,

$$A_i = \frac{1}{2}h(y_{i-1} + y_i), \quad i = 1, 2, \dots, n,$$

so that the total area under f on $[a, b]$ is then

$$A = \sum_{i=1}^n A_i.$$

Now, how accurate is the trapezoidal rule? On each subinterval $[x_{i-1}, x_i]$, $i = 0, 1, \dots, n$, $f(x)$ is replaced by its interpolant using the two nodes x_{i-1}, x_i . Using (6) in Section 2, the error committed on each subinterval is

$$E_i = \int_{x_{i-1}}^{x_i} (f(x) - p_1(x)) dx = \int_{x_{i-1}}^{x_i} \frac{f''(\eta_i)}{2} (x - x_{i-1})(x - x_i) dx,$$

for some $\eta_i \in [x_{i-1}, x_i]$. Now,

$$E_i = \frac{f''(\eta_i)}{2} \int_{x_{i-1}}^{x_i} (x - x_{i-1})(x - x_i) dx = -\frac{f''(\eta_i)}{12} h^3. \quad (60)$$

Exercise 5.1-1: Show that

$$\int_{x_{i-1}}^{x_i} (x - x_{i-1})(x - x_i) dx = -\frac{1}{6} (x_i - x_{i-1})^3.$$

□

This means that if $f(x)$ is a constant or linear function, then the error is zero. This makes sense because then the interpolant perfectly estimates f . Otherwise, the error E_i is not zero but can be estimated using (60). The total error for the rule (59) is the sum of the individual errors E_i , that is,

$$E = \sum_{i=1}^n \frac{-1}{12} f''(\eta_i) h^3 = -\frac{nh^3}{12} f''(\eta),$$

where $\eta = \max\{\eta_i\}$, $i = 0, 1, \dots, n$. For equally spaced nodes of n subintervals,

$$h = \frac{b-a}{n} \implies nh = b-a.$$

Thus the error when we use the trapezoidal estimate is

$$E_t = -\frac{(b-a)h^2}{12}f''(\eta). \quad (61)$$

We can see that as $h \rightarrow 0$, $n \rightarrow \infty$ and $E \rightarrow 0$ at quadratic speed. That is, the trapezoidal rule is *second order (or quadratic)*, meaning it is $\mathcal{O}(h^2)$.

5.2 Simpson's Rule

In the general integration formula (56), if we take $n = 2$ subintervals, that is, 3 nodes, then

$$\int_{x_0}^{x_2} f(x)dx \approx h \left(y_0(2) + \frac{4}{2}\Delta y_0 + \frac{(4)(4-3)}{12}\Delta^2 y_0 \right).$$

Thus

$$\begin{aligned} \int_{x_0}^{x_2} f(x)dx &\approx 2h \left(y_0 + \Delta y_0 + \frac{\Delta^2 y_0}{6} \right) \\ &= \frac{h}{3}(y_2 + 4y_1 + y_0) \end{aligned}$$

Hence

$$\int_{x_0}^{x_2} f(x)dx \approx \frac{h}{3}(y_2 + 4y_1 + y_0). \quad (62)$$

Formula (62) is called Simpson's $\frac{1}{3}$ -rule for approximating the integral of f on $[x_0, x_2]$. The geometrical interpretation of Simpson's $1/3$ rule is illustrated in Figure 5. The function $f(x)$ is replaced with its quadratic interpolant on interval $[x_0, x_2]$ using the three nodes x_0, x_1 and x_2 .

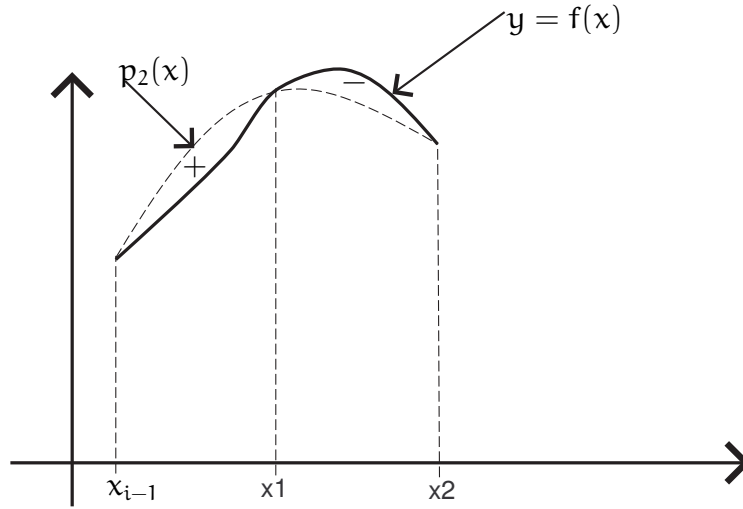


Figure 5: Geometrical illustration of Simpson's 1/3 rule error cancelation property.

In the composite Simpson's rule, the interval $[a, b]$ is subdivided into an even number of subintervals so that (62) is repeated on

$$[x_2, x_4], [x_4, x_6], \dots, [x_{n-2}, x_n],$$

and the integral (52) is the sum of the integrals on these subintervals. So we have

$$\int_a^b f(x)dx \approx \frac{h}{3}(y_0 + 4y_1 + y_2) + \frac{h}{3}(y_2 + 4y_3 + y_4) + \dots + \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n),$$

which simplifies to

$$\int_a^b f(x)dx \approx \frac{h}{3} [y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n]. \quad (63)$$

where n must be even (hence the number of nodes $(n + 1)$ must be odd). Formula (63) is what is known as *Simpson's 1/3-rule* (or most commonly as just *Simpson's rule*).

Exercise 5.2-1: Using the interpolation error formula

$$f(x) - p_2(x) = \frac{f'''(\eta_i)}{24}(x - x_i)(x - x_{i+1})(x - x_{i+2})$$

on each interval $[x_i, x_{i+2}]$ show that the Simpson's third rule on $[a, b]$ will be of higher order than $\mathcal{O}(h^3)$. This higher performance is due to symmetries in integration as is also the case with the midpoint rule. $\square$

If we use three nodes x_0, x_1, x_2 and x_3 in the general integration formula (56), then all differences higher than $\Delta^3 y_0$ are zero and we obtain

$$\int_{x_0}^{x_3} f(x) dx \approx h(3y_0 + \frac{9}{2}\Delta y_0 + \frac{(9)(6-3)\Delta^2 y_0}{12} + \frac{9\Delta^3 y_0}{24})$$

which simplifies to

$$\int_{x_0}^{x_3} f(x) dx \approx \frac{3}{8}h(y_0 + 3y_1 + 3y_2 + y_3). \quad (64)$$

Formula (64) is known as Simpson's $\frac{3}{8}$ th rule for the interval $[x_0, x_3]$.

Exercise 5.2-2: Formulate the general composite Simpson's 3/8 - rule for the interval $[a, b]$. $\square$

Example 5.1 Estimate the integral

$$\int_0^1 \frac{1}{1+x} dx$$

using:

(a) Trapezoidal rule

(b) Simpson's $\frac{1}{3}$ rd rule
with

- (i) $h = 0.5$
- (ii) $h = 0.25$
- (iii) $h = 0.125$
- (c) Using the exact value of the integral, compute the errors in each case and hence comment on the convergence of the method used.
- (d) Use the error formulas (61) to estimate the errors. Compare your results with those obtained in (c).

Solution: With $h = 0.25$, we have the following data:

x	0	0.25	0.50	0.75	1.0
y	1	0.8	0.6667	0.5714	0.5

(a)(i) Using the trapezoidal rule:

$$\begin{aligned}
 I &\approx \frac{h}{2}(y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n) \\
 &= \frac{h}{2}(y_0 + 2(y_1 + y_2 + y_3) + y_4) \\
 &= \frac{0.25}{2}(1 + 2(0.8 + 0.6667 + 0.5714) + 0.5) \\
 &= 0.6970
 \end{aligned}$$

Error is

$$E = |\text{Exact} - \text{Estimate}| = |\ln 2 - 0.6970| = 0.00386$$

(b)(i) Using Simpson's rule:

$$\begin{aligned}
 I &\approx \frac{h}{3}(y_0 + 4(y_1 + y_3 + y_5 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2}) + y_n) \\
 &= \frac{h}{3}(y_0 + 4(y_1 + y_3) + 2(y_2) + y_4) \\
 &= \frac{0.25}{3}(1 + 4(0.8 + 0.5714) + 2(0.6667) + 0.5) \\
 &= 0.6932
 \end{aligned}$$

Error is

$$E = |\text{Exact} - \text{Estimate}| = |0.69314 - 0.6932| = 0.00005.$$

5.3 Midpoint Rule

In this case, on each subinterval on $[x_i, x_{i+1}]$, $f(x)$ is approximated by it's value at the midpoint of the subinterval. That is

$$f(x) \approx f_{i+\frac{1}{2}} = f(x_{i+\frac{1}{2}}) = f(x_i + h/2) = f\left(\frac{x_i + x_{i+1}}{2}\right), \text{ for } x \in [x_i, x_{i+1}].$$

Then,

$$\int_{x_i}^{x_{i+1}} f(x) dx \approx \int_{x_i}^{x_{i+1}} f_{i+\frac{1}{2}} dx = (x_{i+1} - x_i) f_{i+\frac{1}{2}} = h f_{i+\frac{1}{2}} \quad (65)$$

The midpoint rule can be interpreted as the area of the trapezium defined by the tangent of f at $x_{i+\frac{1}{2}}$. The equation of this tangent, is given by

$$\frac{t(x) - f(x_{i+\frac{1}{2}})}{x - x_{i+\frac{1}{2}}} = f'(x_{i+\frac{1}{2}}),$$

or,

$$t(x) = f(x_{i+\frac{1}{2}}) + f'(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}}). \quad (66)$$

When we integrate $t(x)$ on $[x_i, x_{i+1}]$ we have

$$\int_{x_i}^{x_{i+1}} t(x) dx = \int_{x_i}^{x_{i+1}} f(x_{i+\frac{1}{2}}) dx + \int_{x_i}^{x_{i+1}} f'(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}}) dx.$$

The integral of the second term is zero due to symmetry so that we remain with

$$\int_{x_i}^{x_{i+1}} t(x) dx = \int_{x_i}^{x_{i+1}} f(x_{i+\frac{1}{2}}) dx,$$

which is the midpoint rule. To derive the error term, consider the Taylor series of f about $x_{i+\frac{1}{2}}$,

$$f(x) = f(x_{i+\frac{1}{2}}) + f'(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}}) + \frac{f''(x_{i+\frac{1}{2}})}{2}(x - x_{i+\frac{1}{2}})^2 + \cdots.$$

When we integrate, we have

$$\begin{aligned} \int_{x_i}^{x_{i+1}} f(x) dx &= \int_{x_i}^{x_{i+1}} f(x_{i+\frac{1}{2}}) dx + \int_{x_i}^{x_{i+1}} f'(x_{i+\frac{1}{2}})(x - x_{i+\frac{1}{2}}) dx \\ &\quad + \int_{x_i}^{x_{i+1}} \frac{f''(x_{i+\frac{1}{2}})}{2}(x - x_{i+\frac{1}{2}})^2 + \cdots. \end{aligned}$$

The first term is the midpoint rule, the second term evaluates to zero, and hence the error of the midpoint rule on $[x_i, x_{i+1}]$ is

$$E_i = \frac{1}{2}f''(\zeta_i) \int_{x_i}^{x_{i+1}} (x - x_{i+\frac{1}{2}})^2 dx = \frac{1}{24}f''(\zeta_i)h^3, \quad (67)$$

for some $\zeta_i \in [x_i, x_{i+\frac{1}{2}}]$.

Exercise 5.3-1: Verify the result of the integral in (67). □

We see that, although the midpoint uses one function value, it is exact for $f(x)$ linear, like the trapezoidal rule. This is due to the symmetry of the method as witnessed above in the derivation of the error formula. The composite midpoint rule is obtained by repeating (65) on all the n subintervals and then adding. That is,

$$\int_a^b f(x) dx = \int_{x_0}^{x_n} f(x) dx \approx \sum_{i=0}^{n-1} hf_{i+\frac{1}{2}} = h \sum_{i=0}^{n-1} f_{i+\frac{1}{2}}. \quad (68)$$

Formula (68) is what is known as the midpoint rule for (52). To obtain the error of (68), we sum all the contributions from each subinterval $[x_i, x_{i+1}]$ given by (67).

That is we have

$$E_m = \sum_{i=0}^{n-1} \frac{1}{24} f''(\zeta_i) h^3 = \frac{f''(\zeta)}{24} n h^3,$$

where ζ corresponds to $\max\{\zeta_i\} \forall i$. But $nh = (b - a)$ so

$$E_m = \frac{1}{24} f''(\zeta) (b - a) h^2. \quad (69)$$

Comparing (61) and (69), we see that both the midpoint rule and the trapezoidal rule are second order methods. The error in the midpoint rule, E_m , is roughly half the size of the error in the trapezoidal rule, E_t , and of the opposite sign. It looks like there is a linear combination of the two methods that leads to cancellation of the $\mathcal{O}(h^2)$ terms in the truncation error so that we get a higher order approximation. Consider the linear combination

$$\begin{aligned} \frac{1}{3} E_t + \frac{2}{3} E_m &= -\frac{1}{3} \frac{(b-a)h^2}{12} f''(\eta) + \frac{2}{3} \frac{1}{24} f''(\zeta) (b-a) h^2 \\ &= f''(\gamma) (b-a) h^2 \left(-\frac{1}{36} + \frac{1}{36} \right) = 0, \end{aligned}$$

where we have used $f''(\gamma) = \max\{f''(\eta), f''(\zeta)\}$ in both error formulas. Thus this linear combination of the midpoint rule and the trapezoidal rule should give a method of order greater than second.

Exercise 5.3-2: Show that

$$\frac{1}{3} (\text{Trapezoidal rule}) + \frac{2}{3} (\text{Midpoint rule}) = \text{Simpson's rule} + \mathcal{O}(h^k), k > 2.$$

□

Exercise 5.3-3: Compute the integral

$$\int_0^{\pi/2} \sin(x) dx$$

with $h = h_1 = \pi/4$, $h = h_1/2$, $h = h_1/4$, $h = h_1/8$ using

- (a) The midpoint rule,
- (b) The Trapezoidal rule,
- (c) The Simpson's 1/3 - rule.

For each method, tabulate your results in a table of the form

h	I	Error E_h	$E_h/E_{h/2}$

□

6 Techniques for nonlinear equations in one-dimension

We consider solutions for non-linear equations in one dimension. That is, given a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$, we seek a point $x^* \in \mathbb{R}$ such that

$$f(x^*) = 0. \quad (70)$$

The point x^* is called a *root* or *zero* of $f(x) = 0$. The problem of finding such x^* is sometimes called a *root finding problem*.

Theorem 6.1 (Intermediate Value Theorem (IVT)) *If $f \in C[a, b]$ and k is a number between $m = \min_{a \leq x \leq b} f(x)$ and $M = \max_{a \leq x \leq b} f(x)$, then there exists $c \in [a, b]$ such that $f(c) =$*

k.

Corollary 6.1 (Existence of the solution) *If $f \in C[a, b]$ and $f(a)f(b) < 0$, then there exists $z \in [a, b]$ such that $f(z) = 0$.*

Proof

By the IVT if $f(a)f(b) < 0$ then $\implies m = \min_{a \leq x \leq b} f(x) < 0$ and $M = \max_{a \leq x \leq b} f(x) > 0$.

Hence $0 \in (f(a), f(b))$ so we set $k = 0$ in the IVT such that there exists $z \in [a, b]$ with $f(z) = 0$.

Corollary 6.2 (Uniqueness of the solution) *If $f \in C[a, b]$, f is monotonic on $[a, b]$ and $f(a)f(b) < 0$, then there exists a unique $z \in [a, b]$ such that $f(z) = 0$.*

In the sequel we discuss methods of finding such a solution.

6.1 Bisection Method

The idea of the *bisection method* is to find an interval that contains the root and then split the interval into smaller and smaller sections that contain the root. In the end we have a very small interval that contains the root and we may take the middle point of the the interval as the approximation to the root. The steps are summarised in Algorithm 1.

Example 6.1 Use bisection to find the root of $x^2 = 3$ on $[1, 2]$ with tolerance of 0.05.

Solution: To find the root of $x^2 = 3$, we rearrange in the form $f(x) = 0$. That is,

$$x^2 = 3 \implies x^2 - 3 = 0.$$

Therefore we set $f(x) = x^2 - 3$. Then $f(a) = (1)^2 - 3 = -2$ and $f(b) = (2)^2 - 3 = 1$, hence $f(a)f(b) = f(1)f(2) = (-2)(1) = -2 < 0$. So there is a root in $[1, 2]$ The

Algorithm 1 Bisection Algorithm for $f(x) = 0$ **Find:** $[a, b]$ such that $f(a)f(b) < 0$ **Initialisation:** Set $k = 0$, $a_0 = a$, $b_0 = b$, and TOL

```

while  $\frac{|b_k - a_k|}{2} \geq \text{TOL}$  do
   $x_k = \frac{a_k + b_k}{2}$ 
  if  $f(x_k) = 0$  then
    Stop: Solution is  $x_k$ 
  else if  $f(a_k)f(x_k) < 0$  then
     $a_{k+1} = a_k$ ,  $b_{k+1} = x_k$ 
  else
     $a_{k+1} = x_k$ ,  $b_{k+1} = b_k$ 
  end if
   $k \leftarrow k + 1$ 
end while
print  $x_k$ 

```

k	a_k	$\text{sign}(f(a_k))$	b_k	$\text{sign}(f(b_k))$	x_k	$\frac{ b_k - a_k }{2} < 5 \times 10^{-2}$?
0	1		2		1.5	0.5
1	1.5		2		1.75	0.25
2	1.5		1.75		1.625	0.125
3	1.625		1.75		1.6875	0.0625
4	1.6875		1.75		1.71875	0.03125

Table 8: Bisection method iterations for Example 6.1.

iterations are then carried out in the Table 8. So the root (with the given tolerance) is $x_4 = 1.71875$, since the error $0.03125 < 0.05$. Other stop criteria that could be include $|f(x_k)| < \text{TOL}$ or prescribing the maximum number of iterations. Note that $b_0 = b$, $a_0 = a$, $x_0 = (a_0 + b_0)/2$. Suppose we replace b_0 such that

$$b_1 = x_0 = \frac{b_0 + a_0}{2}, \text{ and } a_1 = a_0.$$

Then

$$b_1 - a_1 = \frac{b_0 + a_0}{2} - a_0 = \frac{b_0 - a_0}{2}.$$

If we again on the next iteration replace b_1 , then

$$b_2 = \frac{\frac{b_0 + a_0}{2} + a_0}{2} = \frac{b_0 + 3a_0}{4} \text{ and } a_2 = a_0.$$

Then

$$b_2 - a_2 = \frac{b_0 + 3a_0}{4} - a_0 = \frac{b_0 - a_0}{2^2}.$$

Generally, we can show by induction that

$$b_k - a_k = \frac{b_0 - a_0}{2^k}. \quad (71)$$

This result is used to prove an important theorem on the error convergence of the Bisection method.

Theorem 6.2 (Convergence of the bisection method) *Assume z is the only solution of $f(x) = 0$ in $[a, b]$. If f is a continuous function on $[a, b]$ and $f(a)f(b) < 0$, then*

$$|x_k - z| < \frac{b - a}{2^{k+1}}, \quad k = 0, 1, 2, \dots$$

Proof

Note that for each $[a_k, b_k]$, $x_k = (a_k + b_k)/2$. Therefore, if $z \in (a, b)$, then

$$|x_k - z| < \frac{b_k - a_k}{2}. \quad (72)$$

Then using (71) in (72) completes the proof. $\square$

Theorem 6.2 means that the error at each iteration is at most half the size of the interval. Let $|I_k|$ be the size of the k -th interval. Since at each iteration we take the midpoint as the solution, then $|I_{k+1}| = \frac{1}{2}|I_k|$. So at each iteration the error is approximately halved. That is, if e_k is the error in x_k , then

$$e_{k+1} = \frac{1}{2}e_k.$$

Thus the bisection method has *linear convergence*. This is rather slow and a disadvantage compared to other available methods. The other disadvantage is that we must find a “good” initial interval $[a, b]$ such that $f(a)f(b) < 0$. However, the method also has some advantages: it requires one function evaluation at each iteration and, the more important one, it is guaranteed to converge once we have the initial interval $[a, b]$ such that $f(a)f(b) < 0$.

Example 6.2 Assume the solution of $f(x) = 0$ is in $[0, 2]$. How many iteration steps of the bisection method are required to reduce the error to less than 10^{-4} ?

Solution: After k steps we have x_k . If z is the exact solution, then we want that the error is $e_k = |x_k - z| < 10^{-4}$. Now, we use Theorem (6.2). That is,

$$e_k = |x_k - z| < \frac{b_0 - a_0}{2^{k+1}} < 10^{-4}.$$

Thus

$$\begin{aligned} \frac{2 - 0}{2^{k+1}} < 10^{-4} &\implies \frac{2}{2^{k+1}} < 10^{-4} \implies \frac{2^{k+1}}{2} > 10^4 \implies 2^{k+1} > 2 \times 10^4 \\ &\implies (k+1) \ln 2 > \ln(2 \times 10^4) \implies k+1 > \frac{\ln(2 \times 10^4)}{\ln 2} \end{aligned}$$

$$\implies k + 1 > 14.28771238 \implies k > 13.28771238 \implies k = 14.$$

Thus the minimum number of iterations is 14. $\square$

6.2 Fixed Point Iteration Scheme

In this section we discuss one of the most popular approaches to solving nonlinear equations.

Definition 6.1 (Fixed point) Given $g : \mathbb{R} \rightarrow \mathbb{R}$, a value x such that $g(x) = x$ is called a fixed point of the function g .

That is, x is unchanged when g is applied to it. Many iterative algorithms/schemes for solving nonlinear equations are based on this principle. They are of the form

$$x_{k+1} = g(x_k), \quad k = 0, 1, 2, \dots \quad (73)$$

When used to solve (70), the function g is chosen such that its fixed points are solutions to $f(x) = 0$. Such a scheme is called a *fixed point iteration*. It starts with an initial guess x_0 and then generates a sequence of numbers

$$x_0, x_1, \dots$$

which if the scheme converges, it converges to the fixed point. To convert a root finding problem for (70) to a fixed point scheme, find $g(x)$ such that $g(x) = f(x) + x$, or

$$f(x) = g(x) - x.$$

If x^* is a fixed point of g , then

$$f(x^*) = g(x^*) - x^* = x^* - x^* = 0.$$

Therefore, x^* will be a root of $f(x)$. Note that for a given $f(x)$ there might be more than one ways of choosing $g(x)$ and thus difference fixed point iterative schemes.

Example 6.3 Use a fixed point scheme to find the root of $f(x) = x^2 - 2x - 3$ on $[2, 4]$ using an initial guess of $x_0 = 4$

Solution: Given $f(x) = x^2 - 2x - 3$, the different forms of g include

$$g(x) = \frac{1}{2}(x^2 - 3), \quad g(x) = \sqrt{2x + 3}.$$

Suppose we want use

$$g(x) = \frac{1}{2}(x^2 - 3).$$

Then we have the scheme

$$x_{k+1} = g(x_k) = \frac{1}{2}(x_k^2 - 3), \quad k = 0, 1, 2, \dots \quad (74)$$

The results of several runs of the scheme with $x_0 = 4$ are shown in Table 9(a).

k	x_k	x_{k+1}
0	4	6.5
1	6.5	19.625
2	19.625	191.07
3	191.07	18.25
4	18.25×10^{-3}	

(a)

k	x_k
0	4
1	3.3166
2	3.1037
3	3.0344
4	3.0114
5	3.0038

(b)

Table 9: Fixed point iterative schemes for $g(x) = (x^2 - 3)/2$ in (a) and $g(x) = \sqrt{2x + 3}$ in (b).

The sequence of numbers generated using the scheme (74) shows no convergence. This is therefore a bad choice of g .

Suppose we now choose $g(x) = \sqrt{2x + 3}$, then we have the scheme

$$x_{k+1} = \sqrt{2x_k + 3}, \quad k = 0, 1, 2, \dots \quad (75)$$

The results of this scheme are shown in Table 9(b). This time the scheme is converging and at $k = 5$, we have the approximation to the root is $x_5 = 3.0038$. Note here that $x_4 = 3.0114$ gives the root correct to 1 DP (i.e, one decimal place is fixed already), x_6 gives the root correct to 2 decimal places (i.e, two decimal places are fixed already.)

Note: Use more decimal places in your computations to minimise rounding errors.

Exercise 6.2-1:

- Extend the results in Table 9 with five more iterations.
- Find other forms of g for the problem in Example 6.3 and hence solve the problem using a fixed point scheme from these g s.
- Find all possible forms of $g(x)$ for solving $f(x) = x^2 - x - 2$.

□

6.2.1 Convergence criterion for fixed point schemes

If x^* is a fixed point of g , that is a root of f , then the error at the k^{th} iteration is

$$e_{k+1} = x_{k+1} - x^* = g(x_k) - g(x^*).$$

We want to establish how fast this error goes to zero, that is, how fast does $x_k \rightarrow x^*$. Recall the mean value theorem:

Theorem 6.3 (Mean Value Theorem (MVT)) *If $f(x)$ is defined and continuous on $[a, b]$ and differentiable on (a, b) , then there exists at least a $c \in (a, b)$ such that*

$$g'(c) = \frac{f(b) - f(a)}{b - a}$$

Consider $g(x)$ on the interval $[x_k, x^*]$, by the MVT, there exists $\alpha \in (x_k, x^*)$ such that

$$g(x_k) - g(x^*) = g'(\alpha_k)(x_k - x^*).$$

Thus

$$e_{k+1} = |g'(\alpha_k)|e_k.$$

The value of α_k is not known but, since $g(x^*) = 0$, close enough to x^* we will have $|g(x)| < 1$. Then by starting the iteration close enough to x^* , we can be assured that there is a constant C such that

$$|g'(\alpha_k)| \leq C < 1.$$

Then

$$e_{k+1} \leq Ce_k \leq C^2e_{k-1} \leq C^3e_{k-2} \leq \dots \leq C^{k+1}e_0.$$

That is,

$$e_{k+1} \leq C^{k+1}e_0.$$

But $C < 1 \implies C^k \rightarrow 0$ as $k \rightarrow \infty$. This means that $e_{k+1} \rightarrow 0$ and the sequence x_k , $k = 0, 1, 2, \dots$ converges. Thus we have the following convergence criterion (theorem);

Theorem 6.4 (Convergence of fixed point iterative scheme) *Let $g(x) \in C[a, b]$ (where (a, b) contains a single root of f) be differentiable on (a, b) . If there exists a number such that $|g'(x)| \leq C < 1$ for all $x \in [a, b]$, then g has exactly one fixed point x^* in $[a, b]$ and the sequence (x_k) generated by $x_{k+1} = g(x_k)$, $k = 0, 1, 2, \dots$ with a starting point $x_0 \in [a, b]$ converges to x^* .*

Example 6.4 Consider the problem of Example 6.3 where we had

$$g_1(x) = \frac{1}{2}(x^2 - 3) \text{ and } g_2(x) = \sqrt{2x + 3}$$

We check $|g(x)|$ on the given interval $[2, 4]$.

$$|g_1'(x)| = |x| > 1 \text{ for } x \in [2, 4].$$

Hence the corresponding scheme will diverge. For $g_2(x)$,

$$|g_2'(x)| = \left| \frac{1}{2\sqrt{2x+3}} \right| < 1 \text{ for } x \in [2, 4].$$

Hence the corresponding scheme will converge for any $x_0 \in [2, 4]$. □

The scheme is summarised in Algorithm 2.

Algorithm 2 Fixed point algorithm $x_{k+1} = g(x_k)$

Find: $[a, b]$, $g(x) : f(a)f(b) < 0$, $|g(x)| < 1 \forall x \in [a, b]$

Initialisation: Set $k = 0$, $x_k = x_0$, and TOL

while $|f(x_k)| \geq \text{TOL}$ **do**

$x_{k+1} = g(x_k)$

$k \leftarrow k + 1$

end while

print x_{k+1}

6.2.2 Order of an iteration method, Speed of convergence

Let $x_{k+1} = g(x_k)$ be an iterative scheme. Let x^* be a fixed point of g . Then the error in x_k is

$$e_k = x_k - x^*.$$

That is

$$x_k = x^* + e_k,$$

and

$$x_{k+1} = g(x_k) = g(x^* + e_k). \tag{76}$$

Suppose g is differentiable many times, then we can use the Taylor series expansion of $g(x^* + e_k)$ about x^* in (76) to have

$$x_{k+1} = g(x^* + e_k) = g(x^*) + g'(x^*)e_k + \frac{1}{2}g''(x^*)e_k^2 + \frac{1}{6}g'''(x^*)e_k^3 + \dots \quad (77)$$

Thus we have

$$x_{k+1} - x^* = g'(x^*)e_k + \frac{1}{2}g''(x^*)e_k^2 + \frac{1}{6}g'''(x^*)e_k^3 + \dots \quad (78)$$

Definition 6.2 (Order of convergence) *The exponent of e_k in the first nonvanishing term on the right hand side of the expansion (78) is the order of convergence of the iteration scheme.*

Order of convergence is the measure of the speed of convergence. If $g'(x^*) \neq 0$, then

$$x_{k+1} - g(x^*) = e_{k+1} \approx g'(x^*)e_k,$$

and the method is *first order* or is of order 1. That is

$$\frac{|e_{k+1}|}{|e_k|} = |g'(x^*)| = \text{constant}$$

for a first order method.

If $g'(x^*) = 0$ but $g''(x^*) \neq 0$, then $e_{k+1} = \frac{1}{2}g''(x^*)e_k^2$ and the method is *second order*. That is

$$|e_{k+1}| = C|e_k|^2, \text{ or } \frac{|e_{k+1}|}{|e_k|^2} = \text{constant}$$

for a second order method. A second order method is faster than a first order method, a third order method is faster than a second order method, etc.

Exercise 6.2.2-2: Show that if $|g'(x)| < k$ in a neighbourhood of the root x^* , then

$$|x_{n+1} - x^*| \leq \frac{k}{1-k} |x_{n+1} - x_n|$$

6.3 Newton's method

This is also known as the *Newton-Raphson Method (NRM)*. It is another method for solving $f(x) = 0$ where f is assumed to have a continuous derivative $f'(x)$. The idea is that we approximate the graph of f by suitable successive tangents and each time the solution of the tangent equation is taken as the approximation to the solution of $f(x) = 0$.

Given an approximate value x_0 , we let x_1 be the point of intersection of the tangent to the curve of f at x_0 with the x -axis. Figure 6 gives an illustration.

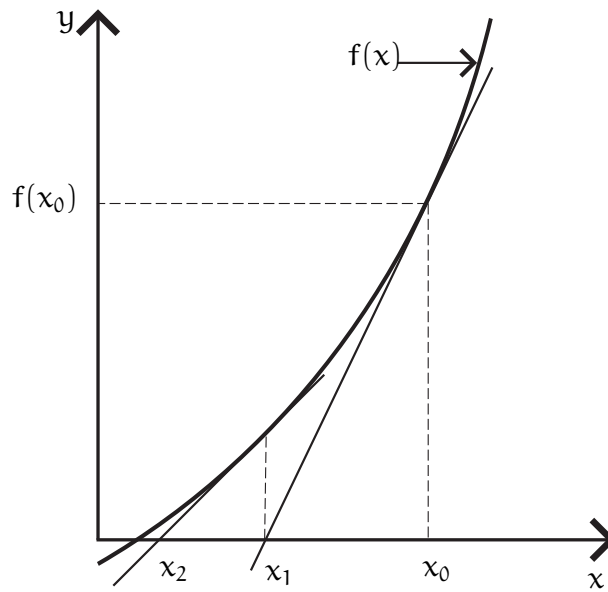


Figure 6: Geometrical illustration of the NRM method.

The equation of the tangent at x_0 is

$$y = f(x_0) + (x - x_0)f'(x_0).$$

At $x = x_1$, $y = 0$ so

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

The process is repeated at $x = x_1$ and so on so that, generally, at $x = x_k$ we get

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \quad k = 0, 1, 2, \dots \quad (79)$$

which is the Newton-Raphson method. The scheme is summarised in Algorithm 3.

Algorithm 3 Newton-Raphson Algorithm for $f(x) = 0$

Initialisation: Set $k = 0$, $x_k = x_0$ (initial guess) and TOL

while $|x_{k+1} - x_k| \geq \text{TOL}$ **do**

 Compute $f'(x_k)$

if $f'(x_k) = 0$ **then**

print failure

 Stop

else

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

end if

$k \leftarrow k + 1$

end while

print x_k

Example 6.5 Set up a Newton-Raphson iteration scheme for computing the positive square root x of a number c and apply it to $c = 2$.

Solution: Let $x = \sqrt{c}$, then $x^2 = c$, so we have

$$x^2 - c = 0.$$

Hence set $f(x) = x^2 - c$. We need $f'(x) = 2x$. Then

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{x_k^2 - c}{2x_k} = \frac{1}{2} \left(x_k + \frac{c}{x_k} \right).$$

Thus the scheme is

$$x_{k+1} = \frac{1}{2} \left(x_k + \frac{c}{x_k} \right), k = 0, 1, 2, \dots$$

Using $x_0 = 1$, the iterations are shown in Table 10.

Example 6.6 Use the NRM to find the positive solution of $2 \sin x = x$

Solution: In this case, $2 \sin x = x$, $\implies x - 2 \sin x = 0$. So we set

$$f(x) = x - 2 \sin x.$$

k	$f(x_k)$	$f'(x_k)$	x_k	$ x_{k+1} - x_k $
0			1	
1			1.5	
2			1.41666667	
3			1.414215686	
4			1.414213562	
5			1.414213562	

Table 10: Newton-Raphson iterations for Example 6.5.

We need $f'(x) = 1 - 2 \cos x$. Then the scheme is given by

$$x_{k+1} = x_k - \frac{(x_k - 2 \sin x_k)}{(1 - 2 \cos x_k)} = \frac{2(\sin x_k - x_k \cos x_k)}{(1 - 2 \cos x_k)}.$$

The iterations with $x_0 = 2$ are shown in Table 11.

k	x_k	$f(x_k)$	$f'(x_k)$	x_{k+1}	error
0	2			1.90100	
1	1.90100			1.89552	
2	1.89552			1.89550	
3	1.89550			1.89549	

Table 11: Newton-Raphson iterations for Example 6.6.

□

Exercise 6.3-1: Solve the following using NRM

- (a) $3x + \sin x - e^x = 0$, on $(0, 1)$. Give solution correct to 6 decimal places.
- (b) $f(x) = x^2 - 4 \sin x$ using $x_0 = 3$. Present your results in a Table clearly indicating the stop criterion used.

Exercise 6.3-2: Prove the following theorem:

Theorem 6.5 (NRM is second order) If $f(x)$ is three times differentiable and $f'(x)$ and $f''(x)$ are not zero at a solution x^* of $f(x) = 0$, then for x_0 sufficiently close to x^* , then the NRM is of second order.

The NRM is very popular because of its quadratic convergence speed. However, it has a number of disadvantages:

1. It is a local method, that is, all convergence properties are proven within the locality of the solution. It converges if the guess is sufficiently close to the solution.
2. It requires two functional evaluations at each iteration. (Name them.)
3. It requires a derivative which may not be readily available.
4. The use of $f'(x)$ in NRM can lead to more problems:
 - (a) If $|f'(x_k)|$ is very small, then x_{k+1} will be far away thus delayed convergence. This is illustrated in Figure 7.

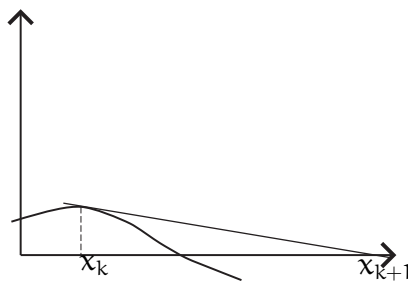


Figure 7: Illustration of effect of small $f'(x_k)$ in the Newton algorithm.

- (b) If the derivative of the function changes sign near x_k , the NRM may oscillate as illustrated in Figure 8.

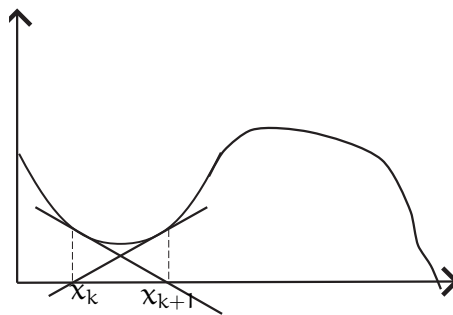


Figure 8: NRM may oscillate if $f'(x)$ is changing signs near the x_k .

6.4 The Secant method

One disadvantage of Newton's method is that $f'(x)$ may not be readily available and even when available, it may sometimes be a more complicated expression than f itself and therefore computationally expensive to evaluate. In the secant method, this is avoided by replacing $f'(x)$ with its finite difference approximation. We use the approximation

$$f'(x_k) \approx \frac{f(x_k) - f(x_{k-1})}{x_k - x_{k-1}},$$

in the NRM algorithm. This leads to

$$x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})} \quad (80)$$

which is the secant method iterative scheme. Geometrically, the secant method is illustrated in Figure 9. In this case, x_{k+1} is the point of intersection of the secant

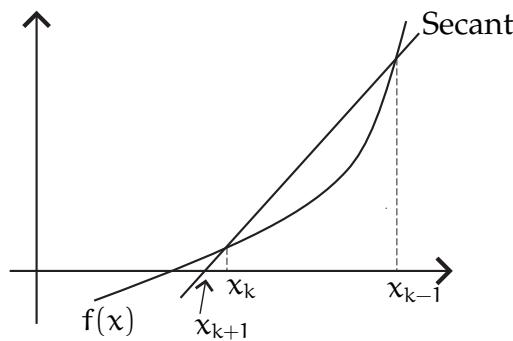


Figure 9: Secant method.

of $f(x)$ passing through $f(x_{k-1})$ and $f(x_k)$ with the x -axis. The Secant method is of the form $x_{k+1} = g(x_k - x_{k-1})$, it requires two previous values to compute the next approximation. It has a number of advantages:

- (1) It requires no evaluation of derivatives.
- (2) Its convergence is super linear, that is, $|e_{k+1}| = C|e_k|^{1.62}$. Better than the bisection method but slightly worse than NRM.
- (3) It can work for double roots.
- (4) It requires only one function evaluation per derivative.

The main disadvantage is that it requires two starting values, x_0 and x_1 .

Exercise 6.4-1:

- (1) Use the secant method to find the square root of 2.
- (2) Find the positive root of $f(x) = x - 2 \sin x$ using the secant method starting with $x_0 = 2$ and $x_1 = 1.9$.
- (3) Use the secant method to find the solution of $f(x) = x^2 - 4 \sin x$ with $x_0 = 1$ and $x_1 = 3$.

Exercise 6.4-2: Prove that the error in the Secant method iterations satisfies

$$|e_{k+1}| = C|e_k|^{1.62}, \quad C \in \mathbb{R}.$$

6.5 The Regular Falsi Method

The Regular-Falsi method, also known as the *method of false position*, is a combination of the secant method and the bisection method. Just as in the bisection method, we begin with a and b for which $f(a)f(b) < 0$. Then the difference is in computing x_k . In this method,

$$x_k = b - \frac{f(b)(b - a)}{f(b) - f(a)}.$$

Next we check the sign of $f(x_k)$. If it is the same as the sign of $f(a)$ then x_k becomes the new a otherwise the new b is x_k and the process is repeated.

Exercise 6.5-1: Find the root of $x^2 - 3 = 0$ on $[1, 2]$ using the regular falsi method to a named tolerance.

7 Techniques for linear systems of equations

We consider the following system of n linear equations in n unknowns.

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, n.$$

Which can be presented in matrix form as

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad (81)$$

where

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

is the coefficient matrix, $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$ is the vector of unknowns and $\mathbf{b} = (b_1, b_2, \dots, b_n)^T$ is the known right hand side. The augmented matrix for the system is

$$\tilde{\mathbf{A}} = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array} \right) \quad (82)$$

In this section we discuss methods of solving such systems other than using the inverse of $\mathbf{A}$ which is an expensive process. The methods are grouped into two categories: *direct methods* and *indirect methods*.

7.1 Direct Methods

Direct methods obtain the solution in a finite number of steps and return an exact solution if all computations are done in infinite precision. The idea is that a triangular system is easy to solve by applying backward substitution (in case of an upper triangular system). How can we then reduce the given system to an equivalent triangular system?

7.1.1 Gaussian Elimination

We assume that the system has a unique solution. The method reduces the system to an upper triangular system using elementary row operations.

Example 7.1 Solve the following using Gaussian Elimination

$$\begin{aligned}x_1 + 2x_2 &= -1 \\2x_1 + x_2 + x_3 &= 3 \\-2x_1 - x_2 + x_3 &= 1\end{aligned}$$

Solution: The augmented matrix is

$$\tilde{A} = \left(\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 2 & 1 & 2 & 3 \\ -2 & -1 & 1 & 1 \end{array} \right)$$

Reducing $\tilde{A}$ to upper triangular gives

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 1 & \frac{-1}{3} & \frac{-5}{3} \\ 0 & 0 & 1 & 2 \end{array} \right).$$

Solving by back substitution gives: $x_3 = 2$, $x_2 = -1$, $x_1 = 1$. Or the solution is $\mathbf{x} = (1, -1, 2)^T$. The gaussian elimination method is to systematically reduce the system (82) into a triangular system through a sequence of finite steps as described below.

Let $A^{(1)}$ denote the initial augmented matrix (82). Re-arrange $A^{(1)}$ so that $a_{11} \neq 0$. If this is not possible, then $\det(A) = 0$ meaning the system is singular. Otherwise we have

$$A^{(1)} = \left(\begin{array}{cccc|c} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} & b_1^{(1)} \\ a_{21}^{(1)} & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} & b_2^{(1)} \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1}^{(1)} & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} & b_n^{(1)} \end{array} \right)$$

Define the $n - 1$ multipliers

$$m_{i1}^{(1)} = -\frac{a_{i1}^{(1)}}{a_{11}^{(1)}}, \quad i > 1.$$

Then reduce $A^{(1)} \rightarrow A^{(2)}$ using the procedure

$$m_{i1}^{(1)} R_1 + R_i \rightarrow R_i, \quad i > 1$$

Example 7.2 From the previous example,

$$m_{21} = -\frac{a_{21}}{a_{11}} = -\frac{2}{1}.$$

□

That is, by elementary row operations, we reduce all the entries below $a_{11}^{(1)}$ to zero so that we have

$$A^{(2)} = \left(\begin{array}{cccc|c} a_{11}^{(1)} & a_{12}^{(1)} & \cdots & a_{1n}^{(1)} & b_1^{(1)} \\ 0 & a_{22}^{(2)} & \cdots & a_{2n}^{(2)} & b_2^{(2)} \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n2}^{(2)} & \cdots & a_{nn}^{(2)} & b_n^{(2)} \end{array} \right)$$

Next define the $n - 2$ multipliers

$$m_{i2}^{(2)} = -\frac{a_{i2}^{(2)}}{a_{22}^{(2)}}, \quad i > 2,$$

and then reduce $A^{(2)} \rightarrow A^{(3)}$ using the operations

$$m_{i2}^{(2)} R_2 + R_i \rightarrow R_i, \quad i > 2.$$

This will generate

$$A^{(3)} = \left(\begin{array}{cccc|c} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1n}^{(1)} & b_1^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2n}^{(2)} & b_2^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3n}^{(3)} & b_3^{(3)} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & a_{n3}^{(3)} & \cdots & a_{nn}^{(3)} & b_n^{(3)} \end{array} \right)$$

Assume $a_{33} \neq 0$, obtain $A^{(4)}$ using the same procedure. Continue until you have

$$A^{(n)} = \left(\begin{array}{cccc|c} a_{11}^{(1)} & a_{12}^{(1)} & a_{13}^{(1)} & \cdots & a_{1n}^{(1)} & b_1^{(1)} \\ 0 & a_{22}^{(2)} & a_{23}^{(2)} & \cdots & a_{2n}^{(2)} & b_2^{(2)} \\ 0 & 0 & a_{33}^{(3)} & \cdots & a_{3n}^{(3)} & b_3^{(3)} \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn}^{(n)} & b_n^{(n)} \end{array} \right)$$

$A^{(n)}$ is upper triangular and $\det(A^{(n)}) \neq 0$.

Definition 7.1 The numbers $a_{11}^{(1)}, a_{22}^{(2)}, a_{33}^{(3)}, \dots, a_{nn}^{(n)}$ are called pivot elements.

The system can now be solved using backsubstitution. That is,

$$x_n = \frac{b_n^{(n)}}{a_{nn}^{(n)}}$$

and

$$x_i = \frac{1}{a_{ii}^{(i)}} \left(b_i^{(i)} - \sum_{j=i+1}^n a_{ij} x_j \right), i = n-1, n-2, \dots, 1.$$

Exercise 7.1.1-1: Solve using Gaussian Elimination:

$$\begin{aligned} x_1 + x_2 + x_3 &= -1 \\ 2x_1 + x_2 + x_3 &= -8 \\ 4x_1 + 6x_2 + 8x_3 &= 14 \end{aligned}$$

Definition 7.2 (Pivoting) The process of swapping two rows to avoid a zero pivot is called pivoting

If the pivot element is so small compared to the elements below it in the same column, we accumulate rounding errors and in the process we may end up with a completely wrong solution. Partial pivoting is used to reduce on the amount of rounding errors when we use Gaussian Elimination.

Definition 7.3 (Partial Pivoting) *The process of swapping rows to ensure that in each column the pivot element is the largest in absolute value of all elements in that column is called partial pivoting.*

Example 7.3 Solving

$$\begin{aligned} 0.003x_1 + 59.14x_2 &= 59.17 \\ 5.291x_1 - 6.130x_2 &= 46.78 \end{aligned}$$

without pivoting using three decimal places yields $x_1 = -10.00$, $x_2 = 1.001$, a wrong solution. Such errors that arise because a computer can only store real numbers with a finite precision are called rounding errors.

7.1.2 Triangular LU Decomposition

Given a system

$$\mathbf{Ax} = \mathbf{b}$$

where $\mathbf{A}$ is a nonsingular matrix, we *decompose/factorise* $\mathbf{A}$ as

$$\mathbf{A} = \mathbf{LU}$$

where $\mathbf{L}$ is *lower triangular* and $\mathbf{U}$ is *upper triangular*. Then the system becomes

$$\mathbf{LUx} = \mathbf{b} \tag{83}$$

For a unique factorisation of $\mathbf{A}$, we fix $\mathbf{L}$ to be *unit lower triangular* (with only ones on the main diagonal) and the resulting triangular decomposition is more specifically known as the Doolittle factorisation. Then, let

$$\mathbf{Ux} = \mathbf{y} \tag{84}$$

so that

$$\mathbf{Ly} = \mathbf{b}. \tag{85}$$

We solve equation (85) by forward substitution to obtain $\mathbf{y}$ and then use $\mathbf{y}$ in equation (84) to obtain $\mathbf{x}$ by backward substitution. If $\mathbf{A}$ is symmetric and positive definite, then we can find the decomposition $\mathbf{LU} = \mathbf{A}$ where $\mathbf{U} = \mathbf{L}^t$. This particular triangular decomposition, that is, $\mathbf{LL}^t = \mathbf{A}$ is the Choleski decomposition of $\mathbf{A}$. Further if $\mathbf{LU} = \mathbf{A}$ where $\mathbf{U}$ is unit upper triangular, then the decomposition is called a Crout decomposition.

Example 7.4 Solve using triangular decomposition

$$\begin{aligned} x_1 + 2x_2 &= -1 \\ 2x_1 + x_2 + x_3 &= 3 \\ -2x_1 - x_2 + x_3 &= 1 \end{aligned}$$

Solution: We have the system

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ -2 & -1 & 1 \end{pmatrix} \mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix}$$

Let $\mathbf{A} = \mathbf{LU}$, that is

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 1 \\ -2 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}.$$

Multiplying out the right hand side gives

$$\mathbf{A} = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{pmatrix}.$$

By equality of matrices, from the first row we get,

$$u_{11} = 1, u_{12} = 2, u_{13} = 0.$$

From the second row,

$$\begin{aligned} l_{21}u_{11} &= 2 \implies l_{21} = 2, \\ l_{21}u_{12} + u_{22} &= 1 \implies u_{22} = 1 - (2)(2) = -3, \\ l_{21}u_{13} + u_{23} &= 1 \implies u_{23} = 1. \end{aligned}$$

From the third row,

$$\begin{aligned} l_{31}u_{11} &= -2 \implies l_{31} = -2, \\ l_{31}u_{12} + l_{32}u_{22} &= -1, \implies l_{32} = -1(-1 + 4)/3 = -1, \\ l_{32}u_{13} + l_{32}u_{23} + u_{33} &= 1, \implies u_{33} = 2. \end{aligned}$$

Hence we have

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -2 & -1 & 1 \end{pmatrix} \text{ and } \mathbf{U} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

Solving $\mathbf{L}\mathbf{y} = \mathbf{b}$, that is,

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -2 & -1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 1 \end{pmatrix},$$

by forward substitution gives $y_1 = -1$, $y_2 = 5$ and $y_3 = 4$.
Then solving $\mathbf{U}\mathbf{x} = \mathbf{y}$, that is,

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & -3 & 1 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 4 \end{pmatrix},$$

by back substitution gives $x_3 = 2$, $x_2 = -1$ and $x_1 = 1$, that is, $\mathbf{x} = (1, -1, 2)^T$.

Exercise 7.1.2-2:

Solve by LU decomposition:

$$\begin{array}{rrrrrr} x_1 & + & 5x_2 & + & 4x_3 & + & 3x_4 & = & -5 \\ 2x_1 & + & 7x_2 & + & 6x_3 & + & 10x_4 & = & 7 \\ 3x_1 & + & 2x_2 & + & 8x_3 & + & 255x_4 & = & 65 \\ 2x_1 & + & x_2 & - & 2x_3 & + & 27x_4 & = & 20 \end{array}$$

7.2 Iterative methods for linear systems

Iterative methods for

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{86}$$

are of the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}, k = 0, 1, 2, \dots \tag{87}$$

The matrix $\mathbf{G}$ is called the *iteration matrix* and the vector $\mathbf{c}$ is called the *iteration vector*. The scheme generates a sequence of vectors $\mathbf{x}^{(k)}$, $k = 0, 1, 2, \dots$, which if converges, it converges to the solution $\mathbf{x}$ of the system. The principal of the schemes discussed here is the same but the difference is in how the iteration matrix $\mathbf{G}$ and vector $\mathbf{c}$ are formulated from $\mathbf{A}$ and $\mathbf{b}$.

7.2.1 Jacobi Iteration Scheme

In this case we decompose $\mathbf{A}$ into

$$\mathbf{A} = \mathbf{L} + \mathbf{D} + \mathbf{U}$$

where $\mathbf{L}$ is the strictly lower part of $\mathbf{A}$, $\mathbf{D}$ is the diagonal of $\mathbf{A}$ and $\mathbf{U}$ is the strictly upper part of $\mathbf{A}$. Then (86) becomes

$$(\mathbf{L} + \mathbf{D} + \mathbf{U})\mathbf{x} = \mathbf{b}. \quad (88)$$

This can be rewritten as

$$(\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{D}\mathbf{x} = \mathbf{b}$$

to give

$$\mathbf{D}\mathbf{x} = -(\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{b}.$$

Hence we have

$$\mathbf{x} = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})\mathbf{x} + \mathbf{D}^{-1}\mathbf{b}.$$

We now set

$$\mathbf{x}^{(k+1)} = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})\mathbf{x}^{(k)} + \mathbf{D}^{-1}\mathbf{b}, k = 0, 1, 2, \dots \quad (89)$$

Equation (89) is in the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}_j\mathbf{x}^{(k)} + \mathbf{c}_j$$

and is the Jacobi iteration scheme where

$$\mathbf{G}_j = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})$$

is the *Jacobi iteration matrix* and

$$\mathbf{c}_j = \mathbf{D}^{-1}\mathbf{b}$$

is the *Jacobi iteration vector*.

Example 7.5 Solve using Jacobi iteration

$$\begin{array}{rrrrr} 10x_1 & + & x_2 & + & x_3 & = & 24 \\ -x_1 & + & 20x_2 & + & x_3 & = & 21 \\ x_1 & + & -2x_2 & + & 100x_3 & = & 300 \end{array}$$

Solution: From

$$\mathbf{A} = \begin{pmatrix} 10 & 1 & 1 \\ -1 & 20 & 1 \\ 1 & -2 & 100 \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} 24 \\ 21 \\ 300 \end{pmatrix},$$

we get

$$\mathbf{L} = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & -2 & 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 10 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 100 \end{pmatrix}, \quad \text{and } \mathbf{U} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

We can now form

$$\mathbf{G}_j = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U}) = -\begin{pmatrix} \frac{1}{10} & 0 & 0 \\ 0 & \frac{1}{20} & 0 \\ 0 & 0 & \frac{1}{100} \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & -2 & 0 \end{pmatrix} = -\begin{pmatrix} 0 & \frac{1}{10} & \frac{1}{10} \\ \frac{-1}{20} & 0 & \frac{1}{20} \\ \frac{1}{100} & \frac{-2}{100} & 0 \end{pmatrix}$$

and

$$\mathbf{c}_j = \mathbf{D}^{-1}\mathbf{b} = \begin{pmatrix} \frac{1}{10} & 0 & 0 \\ 0 & \frac{1}{20} & 0 \\ 0 & 0 & \frac{1}{100} \end{pmatrix} \begin{pmatrix} 24 \\ 21 \\ 300 \end{pmatrix} = \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix}$$

The Jacobi iterations for the problem are then given by

$$\mathbf{x}^{(k+1)} = \begin{pmatrix} 0 & -1/10 & -1/10 \\ 1/20 & 0 & -1/20 \\ -1/100 & 2/100 & 0 \end{pmatrix} \mathbf{x}^{(k)} + \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix}, \quad k = 0, 1, 2, \dots \quad (90)$$

Let

$$\mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

then

$$\mathbf{x}^{(1)} = \begin{pmatrix} 0 & -1/10 & -1/10 \\ 1/20 & 0 & -1/20 \\ -1/100 & 2/100 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix} = \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix},$$

$$\mathbf{x}^{(2)} = \begin{pmatrix} 0 & -1/10 & -1/10 \\ 1/20 & 0 & -1/20 \\ -1/100 & 2/100 & 0 \end{pmatrix} \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix} + \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix} = \begin{pmatrix} 1.995 \\ 1.02 \\ 2.997 \end{pmatrix},$$

$$\mathbf{x}^{(3)} = \begin{pmatrix} 0 & -1/10 & -1/10 \\ 1/20 & 0 & -1/20 \\ -1/100 & 2/100 & 0 \end{pmatrix} \begin{pmatrix} 1.995 \\ 1.02 \\ 2.997 \end{pmatrix} + \begin{pmatrix} 2.4 \\ 1.05 \\ 3 \end{pmatrix} = \begin{pmatrix} 1.9983 \\ 0.9999 \\ 3.0005 \end{pmatrix}$$

Continuing this way we find that the iteration converges to:

$$\mathbf{x} = \begin{pmatrix} 2.000 \\ 1.000 \\ 3.000 \end{pmatrix}.$$

7.2.2 Gauss-Seidel Iteration

In this case, after decomposing (86) into the form (88), we write it as

$$(\mathbf{L} + \mathbf{D})\mathbf{x} = -\mathbf{U}\mathbf{x} + \mathbf{b}.$$

This is rearranged to obtain

$$\mathbf{x} = -(\mathbf{L} + \mathbf{D})^{-1}\mathbf{U}\mathbf{x} + (\mathbf{L} + \mathbf{D})^{-1}\mathbf{b}.$$

We set

$$\mathbf{x}^{(k+1)} = -(\mathbf{L} + \mathbf{D})^{-1}\mathbf{U}\mathbf{x}^{(k)} + (\mathbf{L} + \mathbf{D})^{-1}\mathbf{b}, k = 0, 1, 2, \dots \quad (91)$$

Equation (91) is in the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}_g \mathbf{x}^{(k)} + \mathbf{c}_g$$

and is the Gauss-Seidel iteration scheme, where,

$$\mathbf{G}_g = -(\mathbf{L} + \mathbf{D})^{-1}\mathbf{U}$$

is the *Gauss-Seidel iteration matrix* and

$$\mathbf{c}_g = (\mathbf{L} + \mathbf{D})^{-1}\mathbf{b}$$

is the *Gauss-Seidel iteration vector*.

Example 7.6 Solve the previous example using Gauss Seidel iteration.

Solution:

$$\mathbf{G}_g = -(\mathbf{L} + \mathbf{D})^{-1}\mathbf{u} = \begin{pmatrix} 0 & -1/10 & 1/10 \\ 0 & -1/200 & -11/200 \\ 0 & 9/10000 & -1/10000 \end{pmatrix}$$

and

$$\mathbf{c}_g = (\mathbf{L} + \mathbf{D})^{-1}\mathbf{b} = \begin{pmatrix} 2.4 \\ 1.17 \\ 1.9994 \end{pmatrix}.$$

Let

$$\mathbf{x}^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

then

$$\mathbf{x}^{(1)} = \begin{pmatrix} 0 & -1/10 & 1/10 \\ 0 & -1/200 & -11/200 \\ 0 & 9/10000 & -1/10000 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 2.4 \\ 1.17 \\ 1.9994 \end{pmatrix} = \begin{pmatrix} 2.4 \\ 1.17 \\ 1.9994 \end{pmatrix},$$

$$\mathbf{x}^{(2)} = \begin{pmatrix} 0 & -1/10 & 1/10 \\ 0 & -1/200 & -11/200 \\ 0 & 9/10000 & -1/10000 \end{pmatrix} \begin{pmatrix} 2.4 \\ 1.17 \\ 1.9994 \end{pmatrix} + \begin{pmatrix} 2.4 \\ 1.17 \\ 1.9994 \end{pmatrix}$$

Continuing this way we find that the iteration converges to

$$\mathbf{x} = \begin{pmatrix} 2.000 \\ 1.000 \\ 3.000 \end{pmatrix}.$$

7.3 Convergence of the iteration scheme

Observe that the iterations above are actually fixed point iterative schemes. Thus, If $\mathbf{x}^*$ is the solution then,

$$\mathbf{x}^* = \mathbf{G}\mathbf{x}^* + \mathbf{c}. \quad (92)$$

Subtracting (92) from (87) gives

$$\mathbf{x}^{(k+1)} - \mathbf{x}^* = \mathbf{G}(\mathbf{x}^{(k)} - \mathbf{x}^*). \quad (93)$$

Define

$$\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$$

as the error in $\mathbf{x}^{(k)}$. Then from (93) we have

$$\mathbf{e}^{(k+1)} = \mathbf{G}\mathbf{e}^{(k)} = \mathbf{G}^2\mathbf{e}^{(k-1)} = \mathbf{G}^3\mathbf{e}^{(k-2)} = \dots = \mathbf{G}^{k+1}\mathbf{e}^{(0)}. \quad (94)$$

where $\mathbf{e}^{(0)}$ is the error in the initial guess to the solution. This helps us prove the following theorem

Theorem 7.1 (Convergence if $\mathbf{G}$ is convergent) *The iterative scheme (87) converges to a limit with any arbitrary choice of the initial approximation $\mathbf{x}^{(0)}$ if and only if $\mathbf{G}$ is a convergent matrix.*

Proof:

If $\mathbf{G}$ is a convergent matrix, then $\mathbf{G}^n \rightarrow \mathbf{O}$ as $n \rightarrow \infty$, where $\mathbf{O}$ is the zero matrix. Then using this in (93) gives

$$\mathbf{e}^{(k+1)} = \mathbf{G}^{k+1}\mathbf{e}^{(0)} \rightarrow \mathbf{O}\mathbf{e}^{(0)} = \mathbf{0} \text{ as } k \rightarrow \infty.$$

Hence the scheme converges. On the other hand, since $\mathbf{e}^{(0)}$ is fixed at the beginning

of the scheme, then $\mathbf{G}^{k+1}\mathbf{e}^{(0)} \rightarrow \mathbf{0} \implies \mathbf{G}^{(k+1)} \rightarrow \mathbf{O}$ as $k \rightarrow \infty$. □

The spectral radius of the iteration matrix $\mathbf{G}$ is also fundamental in determining the convergence of the scheme (87).

Theorem 7.2 (Convergence if $\rho(\mathbf{G}) < 1$) *The iterative scheme (87) converges if and only if the spectral radius of $\mathbf{G}$ is less than 1, that is $\rho(\mathbf{G}) < 1$.*

Proof:

Let

$$\mathbf{G}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}$$

be an eigen value decomposition of $\mathbf{G}$, that is, $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues of $\mathbf{G}$ and $\mathbf{V}$ is a matrix whose columns are eigenvectors of $\mathbf{G}$. For convergence, we have that

$$\mathbf{e}^{(k)} = \mathbf{G}^k\mathbf{e}^{(0)} \rightarrow \mathbf{0} \text{ as } k \rightarrow \infty.$$

This means that $\mathbf{G}^k \rightarrow \mathbf{O}$ as $k \rightarrow \infty$ since $\mathbf{e}^{(0)}$ is fixed by the initial guess. Since $\mathbf{G}^k$ has the same eigenvectors as $\mathbf{G}$, to be precise,

$$\mathbf{G}^k \mathbf{V} = \mathbf{V} \mathbf{\Lambda}^k,$$

then $\mathbf{G}^k \rightarrow \mathbf{O}$ if and only if $\mathbf{\Lambda} \rightarrow \mathbf{O}$. This means that $\lambda_i \rightarrow 0$ as $k \rightarrow \infty$ for all i , since $\mathbf{\Lambda}$ is a diagonal matrix of eigenvalues of $\mathbf{G}$, $\lambda_i, i = 1, 2, 3, \dots, n$. That is to say, we have convergence if and only if $|\lambda_i| < 1 \ \forall i$. Thus we have convergence if and only if $\rho(\mathbf{G}) < 1$.

Rate of convergence

Now, suppose $\mathbf{G}$ has n eigenvectors and eigenvalues $\mathbf{V}_i$ and λ_i respectively, $i = 1, 2, 3, \dots, n$. We can use the $\mathbf{v}_i$'s as a basis for $\mathbf{e}^{(0)}$ so that

$$\mathbf{e}^{(0)} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad (95)$$

for some $\alpha_i \in \mathbb{R}, i = 1, 2, \dots, n$. Then

$$\mathbf{e}^{(1)} = \mathbf{G} \mathbf{e}^{(0)} = \mathbf{G} \sum_{i=1}^n \alpha_i \mathbf{v}_i = \sum_{i=1}^n \alpha_i \mathbf{G} \mathbf{v}_i = \sum_{i=1}^n \alpha_i \lambda_i \mathbf{v}_i.$$

Similarly

$$\mathbf{e}^{(2)} = \mathbf{G} \mathbf{e}^{(1)} = \mathbf{G} \sum_{i=1}^n \alpha_i \lambda_i \mathbf{v}_i = \sum_{i=1}^n \alpha_i \lambda_i \mathbf{G} \mathbf{v}_i = \sum_{i=1}^n \alpha_i \lambda_i^2 \mathbf{v}_i.$$

Proceeding in a similar way gives

$$\mathbf{e}^{(k)} = \sum_{i=1}^n \alpha_i \lambda_i^k \mathbf{v}_i. \quad (96)$$

This further shows that $\mathbf{e}^{(k)} \rightarrow \mathbf{O}$ as $k \rightarrow \infty$ if and only if $\lambda_i^k \rightarrow 0$ as $k \rightarrow \infty \ \forall i$. In particular, the behaviour of $\mathbf{e}^{(k)}$ will be dominated by the largest λ_i , that is, if

$$\rho = \max_i |\lambda_i|,$$

then ρ determines the rate of convergence of the scheme. The smaller the value of ρ , the faster the convergence.

Example 7.7 How many iterations does it take to reduce the initial error by a factor 10^{-3} ?

Solution: Let ρ be the spectral radius of $\mathbf{G}$. Then show that

$$\mathbf{e}^{(k)} = \rho^k \mathbf{e}^{(0)} \implies \|\mathbf{e}^{(k)}\| = \rho^k \|\mathbf{e}^{(0)}\|.$$

So after the k^{th} iteration, we want that $\rho^k = 10^{-3}$, $\implies k = \log 10^{-3} / \log \rho$

Definition 7.4 The number $\log \rho$ is called the speed of convergence.

Theorem 7.3 (Convergence due to Diagonal dominance) If $\mathbf{A}$ is strictly diagonally dominant, then the sequence of solutions produced by either Jacobi or Gauss-Seidel iteration converges to the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ for any $\mathbf{x}^{(0)}$.

Proof: A square matrix $\mathbf{A}$ is said to be strictly diagonally dominant if

$$|a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}| \quad \text{for all } i = 1, 2, 3, \dots, n.$$

For Jacobi,

$$\mathbf{G} = -\mathbf{D}^{-1}(\mathbf{L} + \mathbf{U}),$$

then

$$\begin{aligned} \|\mathbf{x}\|_{\infty} &= \frac{\|\mathbf{L} + \mathbf{U}\|_{\infty}}{\|\mathbf{D}\|_{\infty}} \\ &= \frac{\max_{1 \leq i \leq n} \sum_{j=1, j \neq i} |a_{ij}|}{\max_{1 \leq i \leq n} |a_{ii}|} < 1 \end{aligned}$$

since $\mathbf{A}$ is strictly diagonally dominant.

Exercise 7.3-1: Check that

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 1 \\ 1 & -4 & 2 \\ -2 & -1 & 5 \end{pmatrix}$$

is strictly diagonally dominant.

Exercise 7.3-2:

- (a) Consider the iteration scheme (87). Show that $\|\mathbf{e}^{(k+1)}\| \leq \|\mathbf{G}\|^k \|\mathbf{e}^{(0)}\|$.
- (b) Check for the convergence of the Jacobi and Gauss-Seidel schemes for linear systems whose:
- (i)

$$\mathbf{A} = \begin{pmatrix} 3 & 3 & 3 \\ 0 & 2 & 1 \\ 1 & 0 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 6 \\ 5 \\ 4 \end{pmatrix}$$

(ii)

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 1 & 2 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 6 \\ 3 \\ 3 \end{pmatrix}$$

- (c) Verify that the spectral radius for Gauss-Seidel is less than that for Jacobi, that is, $\rho(\mathbf{G}_g) < \rho(\mathbf{G}_j)$. The implication of this is that the Gauss-Seidel scheme converges faster than the Jacobi scheme.

Reading List

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- [5] John H. Mathews and Kurtis D. Fink. *Numerical Methods Using Matlab*. Pearson Education International, New Jersey, 2004.
- [6] Christopher J. Zarowski. *An introduction to numerical analysis for electrical and computer engineers*. Wiley, New Jersey, 2004.

A Past Assignments

A.1 Assignment 2012

(1) Consider a set of $n + 1$ data points

$$(x_k, f_k), \quad k = 0, 1, 2, \dots, n. \quad (97)$$

- (a) (i) Differentiate between what is meant by *interpolation* and an *interpolant*.
 (ii) What is the basic difference between interpolation and approximation?
 (iii) TRUE or FALSE?: There are arbitrarily many different mathematical functions that interpolate a given set of data points.
- (b) A function $P(x)$ that interpolates the data (x_k, f_k) can be expressed as

$$P(x) = \sum_{j=0}^n \alpha_j \phi_j(x)$$

where the parameters α_j are to be determined and ϕ_j s are basis functions. This will result in solving a system of the form

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{f}, \quad \text{where } \mathbf{f} = [f_0 \ f_1 \ f_2 \ \dots \ f_n]^T.$$

In the case of polynomial interpolation, derive the matrix $\mathbf{A}$ (in general form) for the $n + 1$ data points in (102) using

- (i) The monomial basis
 (ii) The Lagrange basis
 (iii) The Newton basis
- (c) Given the data

x	1	2	3	4
f	11	29	65	125

- (i) Determine the polynomial interpolant to the data using the monomial basis.
 (ii) Determine the polynomial interpolant to the data using the Lagrange basis.
 (iii) Determine the Newton polynomial interpolant using

- (I) the matrix method approach
- (II) the divided differences approach
- (iv) Compare your results in (i), (ii) and (iii) above.
- (d) In interpolating discrete data points, we have not worried about the behaviour of the interpolant in between the points. If the data is a sample of an underlying continuous function then this might be important. For polynomial interpolation, there is the following result. If f is a sufficiently smooth function and p_n is the polynomial of degree at most n that interpolates f at $n+1$ points $x_0, x_1, x_2, \dots, x_n$, then for each $x \in [x_0, x_n]$ there is a $\xi \in (x_0, x_n)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n). \quad (98)$$

Since the point ξ is unknown, a bound for the error in (106) is usually considered. Suppose $|f^{(n+1)}(x)| \leq M$ for all $x \in [x_0, x_n]$, and the points are equally spaced such that $h = x_{i+1} - x_i$, $i = 0, 1, 2, \dots, n-1$. Obtain a bound for $\max_{x \in [x_0, x_n]} |f(x) - p_n(x)|$ in terms of M , h and n .

- (e) Consider the following set consisting of the first four Legendre polynomials, $P_0(x)$, $P_1(x)$, $P_2(x)$ and $P_3(x)$,

$$1, \quad x, \quad (3x^2 - 1)/2, \quad (5x^3 - 3x)/2. \quad (99)$$

- (i) Show that the polynomials are orthogonal on $[-1, 1]$ with respect to $w(x) = 1$.
- (ii) The Legendre polynomials satisfy the recurrence relation

$$(k+1)P_{k+1}(x) = (2k+1)xP_k(x) - kP_{k-1}(x)$$

where $P_k(x)$ is the k -th polynomial. Use this relation to find $P_4(x)$ and $P_5(x)$.

- (f) The k -th Chebyshev polynomial of the first kind is defined on $[-1, 1]$ by

$$T_k(x) = \cos(k \arccos(x)).$$

Obtain $T_0(x)$, $T_1(x)$ and use the multiple angle formula for cosines to show that

$$T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x).$$

- (2) (a) Consider $f(x) = x^2 + 2$. Using a step size of $h = 0.2$ on $[1, 2]$, construct a table of forward finite differences for $f(x)$. From the same table give the following

(i) $\Delta f(1.2)$, $\Delta^2 f(1.4)$

(ii) $\nabla f(1.4)$, $\nabla^2 f(1.6)$

(iii) $\delta f(1.5)$, $\delta^2 f(1.4)$

- (b) Let

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$$

be a polynomial of degree n in x . Show that $\Delta^n f(x) = n!h^n a_n$.

- (c) Prove the following finite difference relations:

(i) $E = e^{hD}$ where D is the differential operator d/dx .

(Hint: $Ef(x) = f(x+h)$ then use Taylor's theorem).

(ii) $(1 + \Delta)(1 - \nabla) = 1$

(iii) $\Delta^3 = E^3 - 3E^2 + 3E - 1$

(iv) $\Delta = E - 1$, $\nabla = 1 - E^{-1} = \frac{E - 1}{E}$ and hence show that

$$\nabla^r y_{x+r} = \Delta^r y_x$$

- (3) (a) Compute the Newton forward difference polynomial that interpolates the data:

x	0	1	2	3
y	1	3	7	13

- (b) Let $x_0, x_1, x_2, \dots, x_n$ be a set of $n+1$ equally spaced points such that $x_i = x_0 + ih$, $i = 0, 1, 2, \dots, n$. Let $y_i = f(x_i)$ for all i be the values of $f(x)$ at the points. Let

$$f_n(x) = a_0 + a_1(x-x_n) + a_2(x-x_n)(x-x_{n-1}) + a_3(x-x_n)(x-x_{n-1})(x-x_{n-2}) + \cdots + a_n(x-x_n)(x-x_{n-1}) \cdots (x-x_1). \quad (100)$$

be a polynomial interpolant for the data. By imposing the condition for interpolation on (104) and then expressing the constants a_i , $i = 0, 1, 2, \dots, n$ in terms of backward differences, show that the Newton

backward difference interpolant is given by¹

$$f_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots + \frac{p(p+1)(p+2)\cdots(p+(n-1))}{n!}\nabla^n y_n \quad (101)$$

where $p = (x - x_n)/h$. Hence show that

$$(i) \left. \frac{df(x)}{dx} \right|_{x=x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \dots \right]$$

$$(ii) \left. \frac{d^2f(x)}{dx^2} \right|_{x=x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \dots \right]$$

(c) The population of a town was as given in the table below:

Year	1891	1901	1911	1921	1931
Population('000)	46	66	81	93	101

Estimate the population for the year 1925.

(4) For $f(x) = e^x$ and $f(x) = 2x^2 + x + 1$,

(a) using forward, backward and central differences, estimate $f'(1)$ with (i) $h = 0.1$ (ii) $h = 0.01$ (iii) $h = 0.001$

Discuss your results in terms of accuracy.

(b) Estimate $f''(1)$.

A.2 Assignment I 2013

(1) Consider a set of $n + 1$ data points

$$(x_k, f_k), \quad k = 0, 1, 2, \dots, n. \quad (102)$$

(a) Let $p_n(x)$ be a polynomial of degree at most n that interpolates the data. Show that p_n is unique. [Hint: Let $q_n(x)$ be another different polynomial

¹Equation (105) is good for estimating f and its derivatives at a point near the end of the data.

of degree at most n that interpolates the data, then $r(x) := p_n(x) - q_n(x)$ is a polynomial of degree at most n . Show that r has $n + 1$ roots and that this is a contradiction.]

- (b) A polynomial $p_n(x)$ that interpolates the data (x_k, f_k) can be expressed as

$$p_n(x) = \sum_{j=0}^n \alpha_j \phi_j(x)$$

where the parameters α_j are to be determined and ϕ_j s are basis functions. This will result in solving a system of the form

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{f}, \quad \text{where } \mathbf{f} = [f_0 \ f_1 \ f_2 \ \cdots \ f_n]^T.$$

In the case of polynomial interpolation, derive the matrix $\mathbf{A}$ (in general form) for the $n + 1$ data points in (102) using and compare and contrast the computational costs of using either the monomial basis, the Lagrange basis or the Newton basis.

- (c) Given the data

x	1	2	3
f	11	29	65

Determine the polynomial interpolant to the data using

- (i) the monomial basis,
 - (ii) the Lagrange basis, and
 - (iii) the Newton basis.
 - (iv) Compare your results in (i), (ii) and (iii) above.
- (d) In interpolating discrete data points, we have not worried about the behaviour of the interpolant in between the points. If the data is a sample of an underlying continuous function then this might be important. For polynomial interpolation, there is the following result. If f is a sufficiently smooth function and p_n is the polynomial of degree at most n that interpolates f at $n + 1$ points $x_0, x_1, x_2, \dots, x_n$, then for each $x \in [x_0, x_n]$ there is a $\xi \in (x_0, x_n)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n). \quad (103)$$

Since the point ξ is unknown, a bound for the error in (106) is usually considered. Suppose $|f^{(n+1)}(x)| \leq M$ for all $x \in [x_0, x_n]$, and the points are equally spaced such that $h = x_{i+1} - x_i$, $i = 0, 1, 2, \dots, n-1$. Obtain a bound for $\max_{x \in [x_0, x_n]} |f(x) - p_n(x)|$ in terms of M , h and n .

- (2) (a) Consider $f(x) = \exp(-x)$. Using a step size of $h = 0.2$ on $[0, 1]$, construct a table of forward finite differences for $f(x)$. From the same table give the following

- (i) $\Delta f(0.2)$, $\Delta^2 f(0.4)$
- (ii) $\nabla f(0.4)$, $\nabla^2 f(0.6)$
- (iii) $\delta f(0.5)$, $\delta^2 f(0.4)$

- (b) Let

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

be a polynomial of degree n in x . Show that $\Delta^n f(x) = n!h^n a_n$ and hence that $\Delta^{n+1} f(x) = 0$.

- (c) Prove the following finite difference relations:

- (i) $E = e^{hD}$ where D is the differential operator d/dx .
(Hint: $Ef(x) = f(x+h)$ then use Taylor's theorem).
- (ii) $(1 + \Delta)(1 - \nabla) = 1$
- (iii) $\Delta^3 = E^3 - 3E^2 + 3E - 1$

- (iv) $\Delta = E - 1$, $\nabla = 1 - E^{-1} = \frac{E-1}{E}$ and hence show that

$$\nabla^r y_{x+r} = \Delta^r y_x$$

- (3) (a) Compute the Newton forward difference polynomial that interpolates the data:

x	0	1	2	3
y	1	3	7	13

- (b) Let $x_0, x_1, x_2, \dots, x_n$ be a set of $n+1$ equally spaced points such that $x_i = x_0 + ih$, $i = 0, 1, 2, \dots, n$. Let $y_i = f(x_i)$ for all i be the values of $f(x)$ at the points. Let

$$f_n(x) = a_0 + a_1(x-x_n) + a_2(x-x_n)(x-x_{n-1}) + a_3(x-x_n)(x-x_{n-1})(x-x_{n-2}) + \dots + a_n(x-x_n)(x-x_{n-1}) \dots (x-x_1). \quad (104)$$

be a polynomial interpolant for the data. By imposing the condition for interpolation on (104) and then expressing the constants a_i , $i =$

$0, 1, 2, \dots, n$ in terms of backward differences, show that the Newton backward difference interpolant is given by²

$$f_n(x) = y_n + p\nabla y_n + \frac{p(p+1)}{2!}\nabla^2 y_n + \frac{p(p+1)(p+2)}{3!}\nabla^3 y_n + \dots + \frac{p(p+1)(p+2)\cdots(p+(n-1))}{n!}\nabla^n y_n \quad (105)$$

where $p = (x - x_n)/h$. Hence show that

$$(i) \left. \frac{df(x)}{dx} \right|_{x=x_n} = \frac{1}{h} \left[\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \dots \right]$$

$$(ii) \left. \frac{d^2f(x)}{dx^2} \right|_{x=x_n} = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \dots \right]$$

(4) For $f(x) = e^x$ and $f(x) = 3x^2 + 2x - 1$,

(a) using forward, backward and central differences, estimate $f'(1)$ with (i) $h = 0.1$ (ii) $h = 0.01$ (iii) $h = 0.001$

Discuss your results in terms of accuracy with respect to h .

(b) Estimate $f''(1)$.

A.3 Assignment I 2015

1. Consider a set of $n + 1$ data points (x_k, f_k) , $k = 0, 1, 2, \dots, n$, from a function $f(x)$ where $f_k := f(x_k)$.

(a) Explain the difference between *interpolation* and *extrapolation*.

(b) Let $p_n(x)$ be a polynomial of degree at most n that interpolates the data. Show that p_n is unique.

(c) Show that the Lagrange basis polynomials satisfy

$$\sum_{k=0}^n L_k(x) = 1.$$

²Equation (105) is good for estimating f and its derivatives at a point near the end of the data.

2. Consider the functions

$$f(x) = x + \frac{2}{x} \quad \text{and} \quad g(x) = 4x^3 - 3x + 2.$$

- (a) By using the nodes $x_0 = 0.5$, $x_1 = 1$, $x_2 = 2$ and $x_3 = 2.5$,
- (i) Find the cubic Lagrange interpolant for both functions
 - (ii) Estimate both functions at $x = 1.5$, $x = 1.2$, $x = 2.7$ and $x = 4$ using the interpolants.
 - (iii) By using the exact values compute the errors in each of your estimates in (ii) above. Comment on your answers.
- (b) For the function $f(x)$, construct a table of divided differences for the data and hence
- (i) Using the table, determine the coefficients $\alpha_j, j = 0, 1, 2, 3$, for the Newton polynomial interpolant.
 - (ii) Compute the Newton interpolant $p_3(x)$ using

$$p_3(x) = \alpha_0\phi_0(x) + \alpha_1\phi_1(x) + \alpha_2\phi_2(x) + \alpha_3\phi_3(x)$$

where α_j s are the coefficients determined above and ϕ_j s are the Newton basis polynomials.

- (iii) Use the polynomial to estimate $f(1.5)$ and $f(4)$ and compute the errors in each case.

3. In interpolating discrete data points, we have not worried about the behaviour of the interpolant in between the points. If the data is a sample of an underlying continuous function then this might be important. For polynomial interpolation, there is the following result. If f is a sufficiently smooth function and p_n is the polynomial of degree at most n that interpolates f at $n + 1$ points $x_0, x_1, x_2, \dots, x_n$, then for each $x \in [x_0, x_n]$ there is a $\xi \in (x_0, x_n)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)(x - x_1)(x - x_2) \cdots (x - x_n). \quad (106)$$

Since the point ξ is unknown, a bound for the error in (106) is usually considered.

- (a) Suppose $|f^{(n+1)}(x)| \leq M$ for all $x \in [x_0, x_n]$, and the points are equally spaced such that $h = x_{i+1} - x_i$, $i = 0, 1, 2, \dots, n - 1$. Obtain a bound for $\max_{x \in [x_0, x_n]} |f(x) - p_n(x)|$ in terms of M , h and n .

- (b) Use the formula (106) to estimate the errors in question 2(a)(iii) above. Compare your results with those obtained earlier.

4. Consider Runge's function

$$r(x) = \frac{1}{1+x^2} \quad (107)$$

- (i) Plot $r(x)$ on the interval $[-5, 5]$. (Use MATLAB and submit printed graph.)
- (ii) Using $n+1$ equally spaced nodes on $[-5, 5]$ with $x_0 = -5$ and $x_n = 5$ find Lagrange interpolants $p_4(x)$ and $p_9(x)$ for $r(x)$.
- (iii) On the same axes, plot $r(x)$ and the interpolants $p_4(x)$ and $p_9(x)$. (Use MATLAB and submit printed graph(s)).
- (iv) Comment on the suitability of $p_4(x)$ and $p_9(x)$ as interpolants for $r(x)$. Do you think increasing n will produce a better interpolant for $r(x)$. Explain your answer.
- (v) Plot the derivatives $r'(x)$, $r''(x)$, $r'''(x)$, ... (Use MATLAB and submit printed graphs). Use these graphs to explain the accuracy of the interpolants $p_4(x)$ and $p_9(x)$ as depicted by the graphs in part (iv) above.

B Past Tests

B.1 Test I 2012

1. (a) To find a polynomial $P_n(x)$ that interpolates the data (x_k, f_k) , $k = 0, 1, 2, \dots, n$ involves finding parameters α_j such that

$$P_n(x) = \sum_{j=0}^n \alpha_j \phi_j(x)$$

where ϕ_j s are basis functions. This will result in solving a system of the form

$$\mathbf{A}\boldsymbol{\alpha} = \mathbf{f}, \quad \text{where } \mathbf{f} = [f_0 \ f_1 \ f_2 \ \dots \ f_n]^T.$$

- (i) Give (derivation not required) the matrix $\mathbf{A}$ in terms of x_k , $k = 0, 1, 2, \dots, n$ when the basis is (I) the monomial basis and (II) the Newton basis. [2 marks]
- (ii) Hence, with reasons based on $\mathbf{A}$, decide which interpolant of the monomial and Newton interpolants is cheaper to obtain. [2 marks]

- (b) Find the Lagrange interpolant to the following function at the prescribed interpolating points: [6 marks]

$$f(x) = \exp(2x) \cos(3x), \text{ at } x_0 = 0, x_1 = 0.3, x_2 = 0.6.$$

2. Consider the following table of experimental displacement data

time (t)	0.0	0.1	0.2	0.3	0.4	0.5
distance (f)	0.0	0.003	0.025	0.180	1.192	5.0

- (a) Use the central difference formula to approximate velocity (df/dt) at time 0.15. [6 marks]
- (b) State the error for the central difference formula used. [2 marks]
3. (a) State the midpoint rule and its error for approximating the integral of a function $f(x)$ over an interval $[a, b]$. [3 marks]
- (b) Find the approximation of $\int_0^1 \exp(-x^2) dx$ using the Trapezium rule with two subintervals of equal width. [6 marks]
- (c) Using an error analysis of the midpoint and the trapezium rules explain why the Simpson's $\frac{1}{3}$ rd rule is better than both of them. [3 marks]

B.2 Test II 2012

- (1) Consider the problem of finding the root of $f(x) = x^2 - 2x - 3$ on $[2, 4]$ using a fixed point iteration scheme of the form $x_{k+1} = g(x_k)$. Two possible candidates for $g(x)$ are

$$g(x) = \frac{1}{2}(x^2 - 3) \quad \text{and} \quad g(x) = (2x + 3)^{1/2}.$$

Which one is suitable for the scheme? Give reasons for your answer. [5 marks]

- (2) Assume that the solution of $f(x) = 0$ is in $[0, 2]$ and we are to use Bisection Method to approximate it. What is the minimum number of iteration steps required to reduce the error to less than 10^{-3} ? [5 marks]
- (3) [15 marks]

- (a) Derive a Newton Raphson's iterative scheme for finding the root of $f(x) = x^3 - 3x + 1$.
- (b) Using the scheme in (a) above, find the root on $[0, 1]$ correct to four decimal places using an initial guess of $x_0 = 0.5$. Tabulate your computations.
- (c) Give two disadvantages of Newton Raphson's Method for finding the root of a nonlinear equation.
- (4) If e_k is the error in the k -th iteration and e_{k+1} is the error in the $(k+1)$ -th iteration, compare the Newton Raphson's method and the Secant method in terms of convergence speed and computations. [5 marks]

B.3 Test III 2012

- (1) (a) Explain the following terms as applied in Gaussian elimination technique for linear systems of equations: [3 marks]
 - (i) Pivoting
 - (ii) Partial pivoting
- (b) Solve the following system using Gaussian elimination: [7 marks]

$$x_1 + x_2 + x_3 = 1$$

$$x_1 + x_2 - 2x_3 = 3$$

$$2x_1 + x_2 + x_3 = 2$$

- (2) Use LU-decomposition to solve the following system [10 marks]

$$\begin{pmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ -7 \\ 3 \end{pmatrix}$$

where L is a unit lower triangular matrix.

- (3) Consider solving a linear system

$$A\mathbf{x} = \mathbf{b}$$

using an iterative scheme of the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}$$

- (a) Show that if $\|\mathbf{G}\| < 1$, then the sequence of solutions $\mathbf{x}^{(k)}$, $k = 1, 2, 3, \dots$ generated by the scheme converges for any initial guess $\mathbf{x}^{(0)}$. [4 marks]

- (b) For the Jacobi and Gauss-Seidel iterative schemes, which one is faster? Give a reason. [2 marks]
- (c) Compute the Jacobi iteration matrix for the following system [4 marks]

$$\begin{pmatrix} 3 & 5 & 2 \\ 0 & 8 & 2 \\ 6 & 2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ -7 \\ 3 \end{pmatrix}$$

(Do not perform the iterations.)

B.4 Test I 2013

1. Let $(x_k, f_k), k = 0, 1, 2, \dots, n$, where $f_k = f(x_k)$, be given data for some real valued function $f(x)$. We seek an interpolation polynomial $P_n(x)$ of degree at most n for the data.
 - (i) Explain the advantages, in terms of computation, of using the Lagrange basis over the monomial basis to find $P_n(x)$. [2 marks]
 - (ii) Show that the Lagrange polynomial interpolant $P_n(x)$ is unique. [2 marks]
 - (iii) Find the linear Lagrange polynomial interpolant for $f(x) = e^{-x}$ using the nodes $x_0 = 0, x_1 = 1$. [6 marks]
 - (iv) Explain the difference between extrapolation and interpolation. [2 marks]
 - (v) Explain, in terms of errors, why we would rather use $P_n(x)$ in (iii) above to extrapolate at $x = 1.2$ than at $x = 1.7$. [2 marks]
2. (a) Define the forward difference operator Δ , the backward difference operator ∇ and the shift operator E . [6 marks]
- (b) Prove the following relations
 - (i) $1 + \Delta = E$ [2 marks]
 - (ii) $(1 + \Delta)(1 - \nabla) = 1$ [2 marks]
- (c) Let $f(x) = x^2 + 1$.
 - (i) Use the central difference formula to estimate $f'(x)$ at $x = 2$ using a step size $h = 0.1$ and compute the error E of your estimate. [4 marks]
 - (ii) Without recomputing $f'(2)$, what do you expect the error to be if we use the central difference formula with $h = 0.01$ to estimate $f'(2)$. [2 marks]

B.5 Test II 2013

i. Consider the integral

$$\int_0^1 e^{-x^2} dx. \quad (108)$$

- (a) Suppose we want to approximate the integral (113) using the trapezoidal rule. What is the largest uniform subinterval size h that would ensure the error is not more than 10^{-3} ?
 - (b) Compute the integral in (113) using Simpson's rule with 10 subintervals.
- ii. (a) Let $x_{k+1} = g(x_k)$ define an iteration method. Define what is meant by order of the method.
- (b) In using an iterative method, let e_k denote the error in the k -th iteration approximation. In one method it is found that

$$\lim_{k \rightarrow \infty} \frac{e_{k+1}}{e_k^3} = C, \text{ where } C \text{ is a constant.}$$

What is the order of this method? Explain your answer.

- (c) Show that the Newton-Raphson method for finding the root of a nonlinear equation is a second order method.
- (d) Using $x_0 = 1$ and $x_1 = 0.7$, use the secant method to find the root of

$$e^{-x} - \tan x = 0.$$

Clearly indicate what tolerance you have used.

iii. Consider the problem of solving

$$\mathbf{Ax} = \mathbf{b}. \quad (109)$$

- (a) Define what is meant by $\mathbf{A}$ being strictly diagonally dominant.
- (b) Show that if $\mathbf{A}$ is strictly diagonally dominant then the Jacobi iteration scheme will converge for any initial guess $\mathbf{x}^{(0)}$.
- (c) Suppose in (0d)

$$\mathbf{A} = \begin{bmatrix} 5 & 1 & 2 \\ 1 & 4 & -2 \\ 2 & 3 & 8 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 19 \\ -2 \\ 39 \end{bmatrix}.$$

- (i) Is $\mathbf{A}$ diagonally dominant? Explain your answer.
- (ii) Find the Jacobi iteration matrix and iteration vector and find $\mathbf{x}^{(1)}$ if $\mathbf{x}^{(0)} = [1 \ 1 \ 1]^T$.

B.6 Test I 2014

- 1 (a) Define interpolation and explain two uses of interpolation. [2 marks]
- (b) Let $(x_k, f_k), k = 0, 1, 2, \dots, n$ where $f_k = f(x_k)$ be data given for some real valued function $f(x)$.
 - (i) Give the formulae for generating the Lagrange polynomial basis and the Newton polynomial basis. [2 marks]
 - (ii) Explain the computational advantages of using the Lagrange basis over the monomial basis in finding the polynomial interpolant $P_n(x)$ for the data. [2 marks]
- (c) Given that $f(-1) = 8, f(0) = 3$ and $f(2) = -1$ for some function $f(x)$.
 - (i) Construct a table of divided differences for this data. [3 marks]
 - (ii) Using the table in (i) above, obtain the coefficients $\alpha_0, \alpha_1, \alpha_2$ of the Newton interpolant for the data

$$P_2(x) = \alpha_0\phi_0(x) + \alpha_1\phi_1(x) + \alpha_2\phi_2(x)$$

where $\phi_i(x), i = 0, 1, 2$ are the Newton basis polynomials. [2 marks]

- (iv) Hence find $P_2(x)$ and use it to interpolate f at $x = 0.5$. [3 mark]
- 2 Let $y_i = f(x_i)$ where $x_i = x_0 + ih, i = 0, 1, 2, \dots, n$ and h is the step size.
 - (a) Define the forward difference Δy_i and hence state an expression for the n^{th} forward difference. [2 marks]
 - (b) Prove the following relations on operators
 - (i) $E\nabla = \nabla E = \Delta$, [3 marks]
 - (ii) $(E + E^{-1}) = \Delta\nabla + 2$ where E is the shift (increment) operator. [3 marks]
 - (c) Given that $y = x^2 - 4x + 5$. Using a step size of $h = 0.1$,
 - (i) Construct a forward difference table for $f(x)$ on $[0.0, 0.2]$. [3 marks]

(ii) From the table, determine the central $\delta y_{\frac{1}{2}}$ and hence estimate

$$\frac{dy}{dx} \text{ at } x = 1/2. \quad [2 \text{ marks}]$$

(iii) Compute the error in your estimate in (ii) above. [2 marks]

(iv) What would the error be if h is reduced to $h = 0.01$? Do not recompute. [1 mark]

B.7 Test II 2014

1. (a) Give 2 reasons why we may use
 - (i) Numerical differentiation [2 marks]
 - (ii) Numerical integration [2 marks]
- (b) Given the function $f(x) = 2x^2 - 1$. With a step size of $h = 0.2$,
 - (i) Use an order one method to approximate $f'(5)$. [3 marks]
 - (ii) Approximate $f''(5)$ using the central difference formula. State also the order of the error committed in this approximation. [5 marks]
2. Given that $f(0) = 0$, $f(0.5) = 3.0$, $f(1.0) = 14$ and $f(1.5) = 69$.
 - (a) Construct a forward difference table for the data. [4 marks]
 - (b) Let h be the step size of discretization and, let $p = \frac{x-x_0}{h}$.
 - (i) State the Newton forward difference formula for $\frac{df}{dx}$. [2 marks]
 - (ii) Use the step size of the data given above and the formula in b(i) above to compute $\frac{df}{dx}$ at $x = 0$ and $\frac{df}{dx}$ at $x = 7$. [6 marks]
3. (a) Evaluate, using the midpoint rule, the integral

$$\int_0^1 e^{-x^2} dx$$

using four equal sub-intervals. [4 marks]

- (b) Which one of the Trapezoidal rule and Simpson's $\frac{1}{3}$ rd rule would produce more accurate results as compared to the midpoint rule using the same number of sub-intervals? Explain your answer. [2 marks]
4. Give at least two differences between the Newton Raphson Method and the Secant Method. [3 marks]

END

B.8 Test I 2015

- (a) (a) (i) Explain what is interpolation. [2 marks]
 (ii) Give two uses of interpolation. [2 marks]
- (b) Let (x_k, f_k) , $k = 0, 1, 2, \dots, n$, where $f_k = f(x_k)$, be data given for some real valued function $f(x)$. Explain the computational advantages of using the Newton basis over the monomial basis in finding the polynomial interpolant $p_n(x)$ for the data. [2 marks]
- (c) Consider $f(x) = e^{-x}$ and the nodes $x_0 = 0, x_1 = 0.5, x_2 = 1$.
 (i) Find the interpolant for f using the Lagrange basis and the given nodes. [5 marks]
 (ii) Construct a table of divided differences for f using the given nodes. [3 marks]
 (iii) With the help of the table of divided differences in (ii) above, construct the polynomial interpolant $p_n(x)$ for f using the Newton basis. [4 marks]
 (iv) Compare your results in (i) and (iii) above. [1 marks]
 (v) Using $p_n(x)$ obtained in (iii) above, extrapolate $f(x)$ at a point of your choice. [1 marks]
- (d) The interpolation error in using $p_n(x)$ to approximate $f(x)$ at some point x can be expressed as

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\eta)}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n), \eta \in [x_0, x_n],$$

provided the derivative exists. Using this formula,

- (i) Show that the error at a node is zero. [2 marks]
 (ii) Show that if f is itself a polynomial of degree n , then the interpolant $p_n(x)$ is exact. [2 marks]
- (b) Let $y_i = f(x_i)$ where $x_i = x_0 + ih$, $i = 0, 1, 2, \dots, n$ and h is the step size.
 (a) For $f(x) = xe^x$, $n = 4$ and $x_0 = 0$, construct a table of forward differences for f . [3 marks]
 (b) Using the table, determine the following
 (i) $\Delta f(0.5)$ (ii) $\nabla f(0.75)$ (iii) $\delta f(0.625)$ (iv) $E^{1/2}f(0.125)$ [8 marks]
 (c) Prove the relation $E = (1 - \nabla)^{-1}$. [2 marks]

- (c) The distance y of a particle at a time t is given by $y(t) = t^2 \cos t$, $0 \leq t \leq \pi/2$. We wish to estimate the velocity $v(t) = dy/dt$ using the backward difference formula and the acceleration $a(t) = d^2y/dt^2$ using the central difference formula both at $t = \pi/3$.
- (a) Determine the maximum step size h that will ensure the error in estimating $v(t)$ is not more than 0.001. [4 marks]
- (b) Using a step size of $h = 0.1$, estimate the velocity and acceleration as wished. [6 marks]
- (c) Use the central difference error formula to estimate the error in the acceleration in (b) above. [3 marks]

B.9 Test II 2015

- (a) Consider the problem of evaluating the integral

$$I = \int_a^b f(x) dx \quad (110)$$

using numerical integration.

- (a) Give one reason why we may consider numerical integration. [2 marks]
- (b) Give the error formulas for the midpoint rule and the trapezoidal rule and hence, for a given stepsize h , compare the accuracy of the two methods in evaluating (110). [3 marks]
- (c) Let M_I , T_I , and S_I denote the Midpoint, Trapezoidal and Simpsons 1/3rd rules respectively for evaluating (110) numerically. Show that the linear combination

$$\frac{2}{3}M_I + \frac{1}{3}T_I$$

gives S_I . [3 marks]

- (d) Using Simpson's 1/3rd rule with 8 subintervals, estimate the integral [7 marks]

$$\int_0^1 \frac{1}{1+x^2} dx.$$

- (b) (a) Consider the problem of finding the roots of a nonlinear equation of the form $f(x) = 0$. Give the orders and hence compare the speeds of convergence of the bisection method, the Newton-Raphson method (NRM) and the secant method for solving such an equation. [4 marks]

- (b) Suppose we want to solve the equation

$$x^3 + x - 1 = 0$$

using an iterative scheme of the form

$$x_{k+1} = g(x_k), \quad k = 0, 1, 2, \dots$$

- (i) Show that the scheme whose

$$g(x) = \frac{1}{1 + x^2}$$

converges for any initial guess x_0 . [3 marks]

- (ii) Give one other possible form of g that can be used. [1 marks]

- (iii) Give the Newton-Raphson scheme for solving the equation and hence, using NRM with $x_0 = 1$ and $\text{TOL} = 10^{-6}$, solve the equation. [7 marks]

B.10 Test III 2015

- (a) (a) Differentiate between pivoting and partial pivoting as used in Gaussian elimination. [2 marks]

- (b) Consider the following system that is to be solved using Gaussian elimination

$$\begin{aligned} x + 2y + 3z &= 6 \\ 2x - 3y + 2z &= 14 \\ 3x + y - z &= -2 \end{aligned}$$

- (i) Compute the minors m_{21} and m_{31} . [2 marks]

- (ii) Solve the system using Gaussian elimination. [4 marks]

- (b) A linear system $\mathbf{Ax} = \mathbf{b}$ has

$$\mathbf{A} = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 4 & 5 & 3 & 3 \\ -2 & -6 & 7 & 7 \\ 8 & 9 & 5 & 21 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} -2 \\ -2 \\ -16 \\ -66 \end{bmatrix}$$

and the $\mathbf{L}$, $\mathbf{U}$ factorisation

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 3 & 1 & 0 \\ 4 & 3 & 2 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{U} = \begin{bmatrix} 2 & 3 & 0 & 1 \\ 0 & -1 & 3 & 1 \\ 0 & 0 & -2 & 5 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Solve the system using LU-decomposition. [6 marks]

- (c) The iteration matrix $\mathbf{G}$ of an iterative scheme for solving a linear system of three equations $\mathbf{Ax} = \mathbf{b}$ has eigenvalues given by

$$2/K, 4/K \text{ and } 8/K$$

where K is a positive integer. What is the minimum value of K for the scheme to converge? [3 marks]

- (d) Using $\mathbf{x}^{(0)} = [0, 0]^T$, carry out two iterations of the Jacobi iteration scheme for solving the linear system

$$\begin{aligned} -3x + 8y &= 5 \\ 8x - 12y &= -11 \end{aligned}$$

[6 marks]

- (e) Consider the problem of solving the system $\mathbf{Ax} = \mathbf{b}$ using an iterative scheme of the form $\mathbf{Gx}^{(k+1)} = \mathbf{Gx}^{(k)} + \mathbf{c}$. Let $\mathbf{e}^{(k)}$ be the error in $\mathbf{x}^{(k)}$.

(a) Show that $\mathbf{e}^{(k+1)} = \mathbf{G}^{k+1} \mathbf{e}^{(0)}$. [3 marks]

(b) Show that if $\mathbf{A}$ is strictly diagonally dominant, then the iteration matrix $\mathbf{G}_j$ of the Jacobi iterative scheme for the linear system $\mathbf{Ax} = \mathbf{b}$ has $\|\mathbf{G}_j\| < 1$. Of what importance is this? [4 marks]

C Past Computer Problems Assignments

C.1 Computer problems 2013

Hand in all your computer routines (on a CD) and reports by 1800HRS on April 25th.

- (a) (a) How many zeros does the function $f(x) = \sin(10x) - x$ have? (Hint: Sketching the graph of $f(x)$ would be very helpful.)
- (b) Use a computer algorithm out of the ones discussed in class to find all the zeros of this function. (Hint: You will need a different starting point or initial bracketing interval for each root.)
- (b) For the equation $f(x) = x^2 - 3x + 2 = 0$, each of the following functions yields an equivalent fixed-point problem:

$$g_1(x) = (x^2 + 2)/3,$$

$$g_2(x) = \sqrt{3x - 2},$$

$$g_3(x) = 3 - 2/3,$$

$$g_4(x) = (x^2 - 2)/(2x - 3).$$

- (a) Analyse the convergence properties of each of the corresponding fixed-point iteration schemes for the root $x = 2$ by considering $|g'_i(2)|$, $i = 1, 2, 3, 4$.
- (b) Confirm your analysis by implementing each of the schemes and verifying its convergence (or lack thereof) and approximate convergence rate.
- (c) Implement the bisection, Newton, and secant methods for solving nonlinear equations in one dimension, and test your implementations by finding at least one root for each of the following equations. What termination criterion should you use? What convergence rate is achieved in each case? Compare your results (solutions and convergence rates).
- (a) $x^2 - 2x - 5 = 0$
- (b) $e^x = x$

C.2 Computer problems 2014

- 1 Consider the following boundary value problem:

$$y'' + 2y' + 2y = 0, \quad y(0) = 1, y(\pi/2) = 0.$$

- (a) Obtain the analytic solution to the problem.
- (b) Derive the system of equations to be used to solve the problem using finite differences.
- (c) Write down a MATLAB routine for numerically solving the problem using finite difference for a given step size h and hence plot graphs of the exact solution and the numerical solution together for $h = h_0 = \pi/20$, $h = h_0/2$, $h = h_0/5$.

C.3 Computer problems 2015

(a) Since

$$\int_0^1 \frac{1}{1+x^2} dx = \pi$$

one can compute an approximate value for π using a numerical integration of the given function. Using a computer implementation of the midpoint, trapezoid and Simpson quadrature rules in MATLAB, compute an approximate value for π in this manner for various stepsizes h . Tabulate your results and try to characterise the error as a function of h for each rule. Compare the accuracy of the rules with each other. Is there any point beyond which decreasing h yields no further improvement?

(b) In celestial mechanics, Kepler's equation $M = E - e \sin(E)$ relates the mean anomaly M to the eccentric anomaly E of an elliptical orbit of eccentricity e , where $0 < e < 1$.

(a) Prove that a fixed point iteration using the iteration function

$$g(E) = M + e \sin(E)$$

is locally convergent.

(b) By a computer implementation in MATLAB,

(i) Use the fixed point iteration in part (a) to solve Kepler's equation for eccentric anomaly E corresponding to a mean anomaly of $M = 1$ (radians) and an eccentricity of $e = 0.5$.

(ii) Use Newton's method to solve the same problem.

(iii) Use the bisection method to solve the same problem.

(iv) Use the Secant method to solve the same problem.

(v) Using your results, compare the convergence properties for the methods used above.

(c) Given $f(x) = e^{0.2x}(1 + x^2)$. By implementing the forward, backward and central difference formulas in MATLAB, estimate, for each method the derivatives $f'(x)$ and $f''(x)$ at $x = 2$ using different values of the stepsize h . Tabulate your results and try to characterise the error as a function of h using the exact values of the derivatives.

D Past Examinations

D.1 Exam 2012

Section A (40 Marks)

- (a) Consider the function $f(x) = \frac{1}{x}$. Construct a Lagrange polynomial that interpolates $f(x)$ at

$$x_0 = 1, \quad x_1 = 1.5, \quad x_2 = 1.7$$

[5 marks]

- (b) Let Δ be the forward difference operator and D be the differential operator defined as $Df(x) = \frac{d}{dx}f(x)$. Show that $E = e^{hD}$, for small h , where E

is the shift operator.

[5 marks]

- (c) Consider $f(x) = x^2 + 2x$.

- (a) Using a step size of $h = 0.01$ approximate $f'(x)$ at $x = 2$ using [3 marks]

- (i) the forward difference formula
(ii) the central difference formula

- (b) Use the exact value to compute the corresponding error in (a)(i) and (a)(ii) above. [2 marks]

- (c) Give reasons for the difference in errors in (b) above. [2 marks]

- (d) Estimate the following integral using Simpson's $\frac{1}{3}$ rd rule with four equally sized subintervals [5 marks]

$$\int_0^1 \frac{1}{x^2 + 1} dx.$$

- (e) When using the bisection method, suppose the initial error is $|e_0| = 0.1$. Compute the error $|e_3|$ after the third consecutive iteration. [3 marks]

- (f) (a) Define what is meant by order of convergence of an iterative method. [2 marks]
 (b) Show that the Newton-Raphson iterative method for finding the root of a nonlinear equation $f(x) = 0$ is at least a second order method. [3 marks]
 (g) Show that the secant iterative scheme for finding the r -th root x of a positive number C is

$$x_{k+1} = x_k + \frac{x_{k-1}(x_k^r - C) + Cx_k - x_k^{r+1}}{x_k^r - x_{k-1}^r}.$$

[5 marks]

- (h) Explain the following terms as used in the Gaussian elimination method for systems of linear equations
 (i) Pivoting
 (ii) Partial pivoting [5 marks]

Section B (30 Marks)

- (i) (a) Consider a set of $n + 1$ data points $\{(x_k, f_k); k = 0, 1, \dots, n\}$ where the x_k s are distinct.
 (i) Show that the n -th degree polynomial $p_n(x)$ that interpolates the $n + 1$ data points is unique.
 (ii) State what is meant by divided differences for the data. [5 marks]
 (b) Consider the following population data for the United States at different times t (years) with $t = 0$ being the year 1900:

t (years)	0	10	20	30
Population in millions	76	92	106	123

- (i) Construct a table of divided differences for the data.
 (ii) Using the table, construct the third degree Newton interpolant for the data.
 (iii) Hence, estimate the population in 1926. [10 marks]
 (j) (a) Let $f(x)$ be a real valued function. Let (x_i, y_i) , $i = 1, 2, \dots, n$ be a set of n data points where $y_i = f(x_i)$.
 (i) Define the forward difference operator Δ , the backward difference operator ∇ and the shift operator E in terms of the y_i s.

(ii) Show that

$$E\nabla = \nabla E = \Delta$$

[8 marks]

(b) The first order derivative of a real valued function $f(x)$ at x can be approximated using the Backward difference formula

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}. \quad (111)$$

(i) Use Taylor's theorem to derive the expression for an estimate of the truncation error when h is small.

(ii) Suppose we wanted to use the formula in (111) to estimate $f'(x)$ for

$$f(x) = \frac{1}{3}x^3 + x^2 - 1$$

at some point $x \in [2, 4]$. What is the largest value of h for which the error is not more than 10^{-3} ? [7 marks]

(k) (a) To estimate the integral

$$\int_a^b f(x) dx$$

using the Trapezoidal rule, the interval $[a, b]$ is divided into n subintervals using the points $x_0, x_1, \dots, x_n$ where $x_0 = a$ and $x_n = b$. Then $f(x)$ is estimated on each subinterval $[x_i, x_{i+1}]$, $i = 0, 1, \dots, n-1$, by a linear polynomial interpolant. Show that then the error in the integral on each subinterval is [7 marks]

$$E_i = -\frac{1}{12}h^3 f''(\eta_i) \text{ for some } \eta_i \in [x_i, x_{i+1}], \text{ and } h = |x_{i+1} - x_i|.$$

(b) (i) Using equal subintervals of size $h = 0.25$, use the Trapezoidal rule to estimate the integral

$$\int_0^1 \frac{e^x}{x+1} dx.$$

- (ii) If the size h is reduced by a factor 2, that is halved, by what factor do you expect the error in the estimate to reduce? Clearly explain your answer. [8 marks]

- (l) (a) Use Gaussian elimination to solve the following system

$$3x_1 - 4x_2 + 5x_3 = -1$$

$$-3x_1 + 2x_2 + x_3 = 1$$

$$6x_1 + 8x_2 - x_3 = 35$$

[5 marks]

(b) Let

$$\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}, \quad k = 0, 1, 2, \dots \quad (112)$$

be an iterative scheme for solving a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ where $\mathbf{A}$ is $n \times n$.

- (i) Define a fixed point $\mathbf{x}^*$ of the scheme (112).
- (ii) Suppose $\mathbf{G}$ has n eigenvectors and eigenvalues $\mathbf{v}_i$ and λ_i respectively for $i = 1, 2, \dots, n$. Let $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}^*$ be the error in $\mathbf{x}^k$. Show that

$$\mathbf{e}^{(k)} = \sum_{i=1}^n \alpha_i \lambda_i^k \mathbf{v}_i, \quad \alpha_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n.$$

[6 marks]

- (c) A certain linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$, where $\mathbf{A}$ is $n \times n$, can be solved using two iterative schemes

$$\mathbf{x}^{(k+1)} = \mathbf{G}_1 \mathbf{x}^{(k)} + \mathbf{c}_1, \text{ or } \mathbf{x}^{(k+1)} = \mathbf{G}_2 \mathbf{x}^{(k)} + \mathbf{c}_2, \quad k = 0, 1, 2, \dots$$

The eigenvalues of $\mathbf{G}_1$ are given by

$$\lambda_i = \frac{1}{2i}, \quad i = 1, 2, \dots, n$$

and those of $\mathbf{G}_2$ are given by

$$\lambda_i = \frac{1}{2i+2}, \quad i = 1, 2, \dots, n.$$

Which one of the two schemes would be a better choice to use? Give reasons for your answer. [4 marks]

D.2 Exam 2013

SECTION A (40 marks)

- (a) Consider a set of $n + 1$ data points $\{(x_k, f_k); k = 0, 1, \dots, n\}$ where the x_k s are distinct. Show that the n -th degree polynomial $p_n(x)$ that interpolates the $n + 1$ data points is unique. [5 marks]

- (b) Find the Lagrange linear interpolant for the following data: [5 marks]

x	0	1
$f(x)$	1	0.5

- (c) Let (x_i, y_i) , where $y_i = f(x_i)$, $i = 1, 2, \dots, n$, be a set of n data points for a real valued function f .

- (a) Define the forward difference operator Δ and the backward difference operator ∇ . [2 marks]

- (b) Prove the following identity [3 marks]

$$(1 + \Delta)(1 - \nabla) = 1.$$

- (d) Given $f(x) = x^2 \exp(x) - 1$, use a second order finite difference method to estimate $f'(2)$ using a step size of $h = 0.1$. [5 marks]

- (e) Rank the following methods for finding the root of a nonlinear equation 1st to 3rd from fastest to slowest: Bisection method, Newton Raphson Method and Secant method. [2 marks]

- (f) We want to use the bisection method to find the root of

$$x^2 - 4 \sin(x) = 0$$

using the initial bracket $[1, 3]$.

- (a) Show that this bracket indeed contains a root. [2 marks]

- (b) Carry out three iterations of the bisection algorithm and give the approximation x_3 to the root. [3 marks]

- (g) We would like to use an iterative scheme of the form $x_{k+1} = g(x_k)$ to find the root $\bar{x} = 3$ of the equation

$$x^2 - 5x + 6 = 0$$

using one of the following forms of $g(x)$:

(1) $g_1(x) = \sqrt{5x - 6}$

(2) $g_2(x) = \frac{1}{x}(5x - 6)$

$$(3) \quad g_3(x) = \frac{1}{5}(x^2 + 6)$$

State with reasons which one of the three is the best to use. [5 marks]

(h) Consider the integral

$$\int_0^1 e^x dx. \quad (113)$$

Suppose we want to approximate the integral (113) using the trapezoidal rule. What is the largest uniform subinterval size h that would ensure the error is not more than 10^{-2} ? [4 marks]

(i) We want to solve the system

$$\begin{aligned} -3x_1 + 8x_2 &= 5 \\ 8x_1 - 12x_2 &= -11 \end{aligned}$$

using the Gauss-Seidel Method. Find the Gauss-Seidel iteration matrix G_g and iteration vector c_g . [4 marks]

SECTION B (30 Marks)

(a) Consider the following data from $f(x) = \exp(-x)$

x	0	0.5	1
$f(x)$	1	0.6065	0.3679

- (a) (i) Find the Lagrange quadratic interpolant $P_2(x)$ for f using the data.
 (ii) Using an interpolation error formula, give an upper bound for the error when we use $P_2(x)$ to estimate $f(0.7)$.

[8 marks]

- (b) (i) Construct a table of divided differences for the data.
 (ii) Using the table, find Newton's interpolant for the data.
 (iii) Using your results in (a) and (b) so far clearly, comment on the uniqueness of the polynomial interpolant for the data.

[7 marks]

(b) (a) Consider the function $f(x) = \exp(-2x)$. Using a step size of $h = 0.25$ on $[0, 1]$:

- (i) Construct a table of forward finite differences for f .

(ii) Explain the convergence of the differences in the table.

[7 marks]

- (b) Let $x_0, x_1, \dots, x_n$ be equally spaced points such that $x_i = x_0 + ih$, $i = 0, 1, \dots, n$. Let $y_i = f(x_i)$, $i = 0, 1, \dots, n$ for some function f . Then

$$f_n(x) = y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \dots + \frac{p(p-1)(p-2)\dots(p-(n-1))}{n!}\Delta^n y_0 \quad (114)$$

where $p = (x - x_0)/h$ is Newton's forward difference interpolation polynomial for f . Using the formula in (114), show that the first derivative of f at $x = x_0$ can be estimated as

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{h} \left(\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} + \dots \right)$$

[4 marks]

- (c) For $f(x) = \exp(-2x)$ use your table in (a) and the result in (b) above to estimate

(i) $\left. \frac{df}{dx} \right|_{x=0}$

(ii) $\left. \frac{df}{dx} \right|_{x=0.5}$

[4 marks]

- (c) (a) Use the Trapezoidal rule to evaluate the integral

$$\int_0^1 \frac{1}{x+1} dx$$

using equal subintervals of size $h = 0.5$, $h = 0.25$ and $h = 0.125$. [8 marks]

- (b) Using the exact value of the integral, compute the error in each of the three cases in (a) above and hence show that the trapezoidal method is indeed a second order method. *[5 marks]*
- (c) Would the midpoint rule perform better in terms of reducing the error with h ? Explain your answer. *[2 marks]*

(d) (a) Consider the problem of solving

$$\mathbf{Ax} = \mathbf{b}.$$

using an iterative scheme of the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}.$$

- (i) Define what is meant by $\mathbf{A}$ being strictly diagonally dominant.
- (ii) Show that the scheme converges if $\|\mathbf{G}\| < 1$.
- (iii) Show that if $\mathbf{A}$ is strictly diagonally dominant then the corresponding Jacobi iteration scheme will converge for any initial guess $\mathbf{x}^{(0)}$.

[8 marks]

(b) Suppose we want to solve the system

$$\begin{aligned} 5x_1 - 2x_2 &= 10 \\ -2x_1 + 10x_2 - 2x_3 &= 15 \\ -2x_2 + 15x_3 &= 30 \end{aligned}$$

using Jacobi iterations with an initial guess $\mathbf{x}^{(0)} = [1 \ 1 \ 1]^T$. The corresponding Jacobi iteration matrix has spectral radius 0.327.

- (i) Estimate the minimum number of iterations required to reduce the initial error by a factor 10^{-3} .
- (ii) Carry out one Jacobi iteration to obtain $\mathbf{x}^{(1)}$.

[7 marks]

D.3 Exam 2014

SECTION A (40 marks)

- (a) Without the use of basis polynomials, find the constants a_1 and a_2 such that the polynomial $p(x) = a_1x^2 + a_2x + 1$ is an interpolant for the data [5 marks]

x	0	1	3
$f(x)$	1	3	55

- (b) Construct a table of divided differences for the function $f(x) = \frac{1}{x^2 + 1}$

using three equally spaced nodes on $[0 \ 1]$ with $x_0 = 0$.

[5 marks]

- (c) Given that a is a constant and h a small step size, show that, for $f(x) = e^{ax}$,

$$\Delta^n f(x) = (e^{ah} - 1)^n e^{ax},$$

where Δ is the forward difference operator. [5 marks]

- (d) Consider $f(x) = \sin x$. Suppose we want to use the central difference formula to estimate $f'(x)$ at some x , determine the maximum step size h that will ensure the error is not more than 10^{-2} . [5 marks]

- (e) Consider the problem of solving $f(x) = 0$ using a fixed point iterative scheme of the form

$$x_{k+1} = g(x_k), \quad k = 0, 1, 2, \dots,$$

where $g(x) \in C[a, b]$ on some interval $[a, b]$ containing the root of f . Show that if $|g'(x)| \leq c < 1$ on $[a, b]$, then the scheme converges for any $x_0 \in [a, b]$. [5 marks]

- (f) Use Simpson's 1/3rd rule with 5 subinterval to estimate the integral [5 marks]

$$\int_0^1 \frac{1}{1+x} dx.$$

- (g) Use Gauss elimination with partial pivoting to solve the following system: [5 marks]

$$\begin{aligned} 2x_1 + 2x_2 + x_3 &= 1 \\ 4x_1 + 2x_2 + 3x_3 &= 2 \\ x_1 + x_2 + x_3 &= 3 \end{aligned}$$

- (h) Consider the system of equations

$$\begin{bmatrix} 1 & -a \\ -a & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

where a, b_1 and b_2 are real constants. For what values of a will the Jacobi iteration for the system converge? [5 marks]

SECTION B (30 Marks)

- (a) (a) The problem of finding a polynomial interpolant $p_n(x)$ for the set of $n + 1$ data points $\{(x_k, f_k); k = 0, 1, \dots, n\}$ where the x_k s are distinct hinges on finding parameters α_j such that

$$p_n(x) = \sum_{j=0}^n \alpha_j \phi_j(x),$$

where $\phi_j(x)$, $j = 0, 1, 2, \dots, n$ are the basis polynomials to be used.

- (i) State the Lagrange basis for polynomial interpolation.
 (ii) Show that in the case of the Lagrange basis, $\alpha_j = f_j$, $j = 0, 1, 2, \dots, n$.
 [6 marks]

- (b) Consider the following data

x	0.1	0.2	0.3	0.4	0.5
$f(x)$	1.40	1.56	1.76	2.00	2.28

The Newton forward difference interpolation polynomial for data of size n is given by

$$p(x) = A_0 + A_1(x-x_0) + A_2(x-x_0)(x-x_1) + \dots + A_n(x-x_0)(x-x_1) \dots (x-x_{n-1}).$$

For $n = 4$, construct the table of forward differences for the data.

- (i) With the help of the table, determine the coefficients $A_0, A_1, \dots, A_4$.
 (ii) Hence find the Newton interpolant for the data.

[9 marks]

- (b) (a) Suppose, we want to compute the integral

$$\int_0^1 e^x dx$$

using the trapezoidal rule with uniform subintervals.

- (i) Find the subinterval size h that will ensure the error is not more than 10^{-3} .
 (ii) Use $h = 0.2$ to estimate the integral using the trapezoidal rule.
 (iii) Compute the error in your estimate in (a)(ii) using the exact value.

[8 marks]

(b) Given that $y(x) = x^2 \exp(x)$.

- (i) With a step size of $h = 0.1$, use the central difference formula to estimate $y''(2)$.
- (ii) Compute the error in your estimate in part (b)(i) above.
- (iii) Without recomputing the derivative, what would the error be if h is reduced to $h = 0.01$? Justify your answer.

[7 marks]

(c) (a) Consider a fixed point iterative scheme of the form $x_{k+1} = g(x_k)$, $k = 0, 1, 2, \dots$

- (i) Define what is meant by order of convergence.
- (ii) Show that the Newton Raphson Method is a second order method.
- (iii) If an estimate x_k using the Newton Raphson method has 2 correct decimal places, how many decimal places will be correct in estimate x_{k+3} . Justify your answer.

[8 marks]

(b) Use four iterations of the Newton Raphson method to estimate the value of $(17)^{1/3}$ using initial approximation $x_0 = 1$.

[7 marks]

(d) Consider the system of equations

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ 4x_1 + 3x_2 - x_3 &= 6 \\ 3x_1 + 5x_2 + 3x_3 &= 4 \end{aligned} \tag{115}$$

(a) Solve the system using LU-decomposition. [8 marks]

(b) An iterative scheme for solving the system (115) has the form

$$\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}, \quad k = 0, 1, 2, \dots$$

- (i) Suppose $\mathbf{G}$ has a spectral radius of 0.2. Estimate the number of iterations required to reduce the initial error by a factor 10^{-2} .
- (ii) Determine $\mathbf{G}$ if the scheme is a Jacobi iterative scheme.

[7 marks]

D.4 Exam 2015

SECTION A (40 marks)

1. Suppose the polynomial $y(x) = a_0x^2 + a_1x + 1$ interpolates the curve $f(x) = e^x$ at the nodes $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, find the value of the coefficients a_0 and a_1 . [5 marks]

2. Suppose $p_n(x)$ is an n -degree polynomial interpolant for a function $f(x)$. The error in using $p_n(x)$ to approximate $f(x)$ at some point x is given by

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\eta)}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n), \eta \in [x_0, x_n],$$

provided the derivatives exist. Using this formula,

- (i) Show that the error at a node is zero. **[2 marks]**
 - (ii) Show that if $f(x)$ is itself a polynomial of degree n , then the interpolant $p_n(x)$ is exact. **[3 marks]**
3. If computed exactly, the following integral yields

$$\int_0^1 \frac{4}{1+x^2} dx = \pi.$$

- (i) Using 4 equally sized subintervals, estimate the value of π from this integral using the Trapezoidal rule. **[4 marks]**
 - (ii) What is the error in your estimate? **[1 mark]**
4. (i) Consider $f(x) = x^2 e^{2x}$. Using $h = 0.2$, compute $\delta f(1)$. **[2 marks]**
- (ii) The forward difference formula for estimating the derivative of a function f at x is

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

where h is the step size. Show that this formula is of order one. **[3 marks]**

5. Consider $f(x) = e^x - 3x - 1$.

- (i) Show that f has a zero in the interval $(1, 3)$. **[2 marks]**
 - (ii) Let x_k be the estimate to the root after k iterations of the bisection algorithm, $k = 0, 1, 2, \dots$, find x_2 . Present your iterations in a table. **[3 marks]**
6. (i) What modification do we make on the Newton-Raphson method to obtain the secant method? **[2 marks]**

- (ii) Formulate the iterative scheme based on the secant method for finding the positive root, $\sqrt{N}$, of a positive real number N . [3 marks]
7. Solve the following system using Gauss elimination [5 marks]
- $$\begin{aligned}x_1 + x_2 + x_3 &= 6 \\3x_1 + 3x_2 + 4x_3 &= 20 \\2x_1 + x_2 + 3x_3 &= 13\end{aligned}$$
8. Consider the scheme $\mathbf{x}^{k+1} = \mathbf{G}\mathbf{x}^k + \mathbf{c}$ for solving a linear system of equations $\mathbf{Ax} = \mathbf{b}$.
- (i) Define what is meant by $\mathbf{G}$ being a convergent matrix. [1 marks]
- (ii) Show that the scheme is convergent if and only if $\mathbf{G}$ is a convergent matrix. [4 marks]

SECTION B (Answer two questions for 30 Marks)

1. We want to interpolate the function $f(x) = xe^x$ on the interval $[0, 1]$ using the nodes $x_0 = 0, x_1 = 0.5$ and $x_2 = 1$.
- (a) (i) Find the quadratic Lagrange interpolant $p_2(x)$ for f using the given nodes. [4 marks]
- (ii) State a bound for the error when $p_2(x)$ is used for interpolation at any $x \in [0, 1]$. [2 marks]
- (b) Using the same nodes as in (a) above,
- (i) find the divided differences $f[x_0]$, $f[x_0, x_1]$ and $f[x_0, x_1, x_2]$. [4 marks]
- (ii) find Newtons polynomial interpolant for f on $[0, 1]$ using your results in b(i) and the given nodes. [4 marks]
- (c) Which of the two interpolants obtained in (a) and (b) is a better interpolant? and why? [1 marks]
2. (a) (i) Let $y(x)$ be a real-valued function of $x \in \mathbb{R}$ and h a small positive step size. Define the forward difference $\Delta f(x)$. [2 marks]
- (ii) Let $f(x) = x^{(n)}$ be the factorial polynomial defined as
- $$x^{(n)} = x(x-h)(x-2h) \cdots (x-(n-1)h)$$
- where n is a positive integer and $h > 0$ is the a step size. Show that $\Delta^n f = n!h^n$. [4 marks]

- (b) Given $f(x) = e^x(1 + x^2)$, use second order finite difference methods to estimate
- (i) $f'(1)$, [4 marks]
 - (ii) $f''(1)$, [4 marks]
- using a step size of $h = 0.2$.
- (c) State the error formulas for the methods used in (b). [2 marks]
3. (a) Use Simpson's third rule to compute the following integral using 4 subintervals [5 marks]

$$\int_0^{0.4} xe^{-x^2} dx.$$

- (b) (i) An iterative method for solving a nonlinear equation $f(x) = 0$ has the following errors at successive iterations: 10^{-2} , 10^{-4} , 10^{-8} , 10^{-16} , ... What is the order of the method? Give a mathematical defence of your answer. [2 marks]
- (ii) The problem of finding the positive square-root x of a positive real number N is equivalent to finding the root of the equation

$$f(x) = x^2 - N = 0$$

using a fixed point iterative scheme of the form $x_{k+1} = g(x_k)$, $k = 0, 1, 2, \dots$. What is the corresponding $g(x)$ if the scheme is a Newton iterative scheme? [4 marks]

- (c) We want to solve the equation $2x - \cos x = 0$ using an iterative scheme of

the form $x_{k+1} = g(x_k)$, $k = 0, 1, 2, \dots$ where $g(x) = \frac{1}{2} \cos x$ and $x_0 = 1$.

- (i) Show that this $g(x)$ gives a convergent scheme. [1 marks]
- (ii) Using a tolerance of 10^{-2} solve the equation using the scheme. Tabulate your iterations. [3 marks]

4. Consider the following linear system of equations

$$\begin{aligned} 5x_1 - 2x_2 &= 18 \\ -2x_1 + 10x_2 - 2x_3 &= -60 \\ -2x_2 + 15x_3 &= 128 \end{aligned}$$

We want to solve the system using an iterative scheme of the form $\mathbf{x}^{(k+1)} = \mathbf{G}\mathbf{x}^{(k)} + \mathbf{c}$.

- (a) Give formulas for $\mathbf{G}$ and $\mathbf{c}$ for the system $\mathbf{Ax} = \mathbf{b}$ if the scheme is
- (i) Jacobi [2 marks]
 - (ii) Gauss-Seidel [2 marks]
- (b) Form the matrix $\mathbf{G}$ and the vector $\mathbf{c}$ if the scheme is a Jacobi scheme. [4 marks]
- (c) Compute the infinity norm of $\mathbf{G}$ and hence comment on the convergence of the scheme. [2 marks]
- (d) Using $\mathbf{x}^{(0)} = [1 \ 1 \ 1]$, carry out two Jacobi iterations to find $\mathbf{x}^{(2)}$. [4 marks]
- (e) If the exact solution is $(2, -4, 8)^\top$, what is the error in $\mathbf{x}^{(2)}$ [1 marks]